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Estimating the Expected Pay-out of Earnout Contracts in Private Acquisitions

KTH Master Thesis Report

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Abstract

The growth of private equity, as well as consolidation trends across other industries, have produced a strong and vibrant mergers and acquisitions market. A challenge during these acquisitions is information asymmetry, which makes agreeing on the transaction price a challenge. An increasingly popular instrument to get around this problem is to use earnout contracts, which puts the difference between what the buyer is willing to pay and what the seller is willing to accept as contingent on future performance of the company. This thesis focuses on testing four different models for estimating the expected pay-out of earnout contracts. The investigated models were geometric Brownian motion, autoregressive integrated moving average, artificial neural network and a hybrid model to forecast the underlying metrics which were used with Monte Carlo methods to compute the expected pay-out of the earnout contract. Furthermore, a bankruptcy adjusted and a model using implied market volatility were evaluated. The results were that the hybrid model showed the most promising predictions when estimating the expected pay-out. The bankruptcy adjustment was not successful since the model failed to reach sufficient accuracy. Using implied market volatility showed inconclusive results.

Keywords: Earnout Contracts, Valuation, Mergers & Acquisitions, Private Equity, Monte Carlo Simulation, Contingent Considerations

Sammanfattning

Tillväxten för riskkapital-industrin och konsolideringstrender inom andra industrier har resulterat i en aktiv marknad för bolagsförvärv. En tydlig utmaning under ett förvärv är informationsasymmetri, vilket gör det svårt att komma överens om bolagets värdering. En alltmer vanlig metod för att lösa detta problem är att använda en tilläggsköpeskilling. Ett sådant kontrakt placerar skillnaden mellan vad köparen är villig att betala och vad säljaren är villig att acceptera som en option baserad på bolagets framtida prestation. Detta examensarbete fokuserade på att testa fyra olika modeller för att skatta den framtida utbetalningen från tilläggsköpeskillingar. De utvärderade modellerna var baserade på geometrisk brownsk rörelse, autoregressive integrated moving average, artificiellt neuralt nätverk och en hybridmodell vilka användes för att generera prediktioner för optionernas underliggande mått. Dessa användes sedan för att med hjälp av Monte Carlo simulering skatta den förväntade utbetalningen från tilläggsköpeskillingen. Utöver detta testades en modell med justering av konkursrisk samt en modell baserad på implicerad volatilitet från börsnoterade optioner. Resultaten visade att hybridmodellen gav bäst prediktioner av den förväntade utbetalningen. Den konkursjusterade modellen påvisade inga signifikanta resultat då den ej nådde tillräckligt hög prediktionsförmåga. Användningen av implicerad marknadsvolatilitet gav ingen tydlig och statistiskt signifikant förbättring.

Nyckelord: Tilläggsköpeskilling, Värdering, Bolagsförvärv, Black-Scholes, Monte Carlo Simulering, Optioner

Contents

Mathematical Notation

d - The acquired company's probability of default

μ - Expected growth or drift in the underlying metric of an earnout contract

n - Sample/data size

π - Price function for an option or other derivative

r - Interest rate

σ^2 - Volatility of the underlying metric of the earnout contract

T - Maturity for the earnout contract

W - Wiener process

$\mathbf{X}_t$ - Stochastic variable for the underlying metric at time t

$\mathbf{x}_t$ - Individual outcome of the underlying process at time t

Z - Refers to the standard normal distribution if nothing further is specified

Word List

Acquisition - When a company buys another company

ANN - Artificial neural network

ARIMA - Autoregressive Integrated Moving Average

CAGR - Cumulative aggregate growth rate

CAPM - Capital Asset Pricing Model

Discount Rate - Rate of return used to discount future cash flows and payments back to their present value

EBIT - Earnings Before Interest and Tax

EBITDA - Earnings Before Interest Taxes Amortization and Depreciation

EC - Earnout Contract

GAAP - Generally Accepted Accounting Principles

Hybrid Model - Combination of an ARIMA and an ANN models

M&A - Mergers & Acquisitions

Merger - An agreement that unites two existing companies into one new company

Private Equity - Investments in companies that are not listed on a stock exchange, i.e., public

Real Option - Options on an underlying asset that is not a traded security

Revenue - The sales a company generates over a period

Risk-free Rate - Theoretical rate of return of an investment in an asset without any risk

1 Introduction

1.1 Introducing the Subject Area

When a company is bought in a mergers and acquisition (M&A) transaction, the company is rarely all paid for upfront in cash [U. Axelson, 2009]. In recent years, it has become increasingly popular to set a share of the deal value as contingent on the future performance of the company after the acquisition, also known as an earnout contract (referred to as EC henceforth) [Battauz et al., 2021]. In other words, the seller of the business receives a part of the transaction value up-front, and part of the payment as an EC contingent on the sold company reaching certain future milestones [U. Axelson, 2009].

ECs are commonly used in acquisitions of private companies with a small number of owners [Grønli and Hussein, 2021]. The purpose is to "agree to disagree" in terms of valuation and still enable a transaction. Furthermore, it gives advantages such as to "lock in" key managers to the company after the acquisition during the critical transition period [Bruner and Stiegler, 2008]. The valuation of an EC is far from trivial. This is partly because the main reason for using an EC is because the buyer and seller cannot come to an agreement on the value of the company [Grønli and Hussein, 2021]. Furthermore, accounting rules mandates that ECs are accounted for using "fair value accounting" [Cadman et al., 2014]. Fair value accounting means that an EC should be accounted for as a liability on the balance sheet of the company issuing the EC at today's fair market value [Ferguson et al., 2021].

The research on how to value ECs is still in its infancy [Battauz et al., 2021]. ECs can have widely different pay-out structures and can be based on different performance metrics [Grønli and Hussein, 2021]. When the pay-out structure is multiple based without option structure there are "best practices" for how to value ECs [Chamberlain and Elkounovitch, 2019]. However, when the pay-out structure is option based one must use option pricing methods to value the contract. The option valuation approach required in these instances is more complicated and is the focus of this thesis. When it comes to valuing ECs with a pay-out structure like a call option, there are two common methodologies. Either one can use an adaptation of the Black Scholes formula or Monte Carlo simulation. However, using either Black-Scholes or Monte Carlo simulation of the underlying financial metric requires estimations of input parameters. The process of estimating these parameters is not standardized and seldom discussed in detail within the research literature [Battauz et al., 2021]. The research to date focuses mostly on systematic estimation of the risk premium, while other parameters are estimated based on historical statistics or expert forecasts, relying on the subjective judgement of industry professionals [Chamberlain and Elkounovitch, 2019].

1.2 Previous Research

Research on the valuation of ECs is limited. Chamberlain and Elkounovitch [2019] published the first large scale report on best practices for valuing ECs. The report gives a holistic view of different valuation challenges. However, the foundation fails to give detailed accounts of how to estimate the future volatility in the underlying metrics, and completely omits any descriptions on how to estimate future growth. Furthermore, the report recommends using Black-Scholes, without doing any justifications for the new characteristics and assumptions for the underlying metric. Arzac [2004] gives an in-depth description of ECs and the valuation process. However, when valuing ECs, the book mostly focuses on linear non-option structures. When it comes to ECs with structures that require option pricing methodology, it is simply stated that "they can often be valued as a call option using the Black-Scholes model". Bruner and Stiegler [2008] has produced a technical note on the valuation of ECs which is often cited in the literature. The article suggests a simple Monte Carlo based valuation methodology for ECs with call option properties, involving the forecasting of potential future scenarios and fitting the corresponding distribution.

The sources above are to a considerable extent focused on the broader questions of EC structuring and their explanations of the actual valuation models and parameter estimations necessary are limited and often omitted.

Tallau [2009] presents a Black-Scholes based valuation approach for ECs in his article "The Value of Earn-out Clauses: An Option-Based Approach". The methodology is based on an adaptation of the Black-Scholes model which allows for valuation of different contract structures. However, the actual parameter estimation is left outside the scope of the article. Grant [2011] explores both applications of Monte Carlo simulation and an adapted Black-Scholes model for the valuation of ECs. The article gives a derivation of the adapted Black-Scholes model and suggest a Monte Carlo approach where the underlying metric is modelled as a Geometric Brownian Motion (GBM). Furthermore, Battauz et al. [2021] has attempted valuing non-linear ECs while taking the default risk of the issuer and the risk of legal consequences into account. The study models both the value of the EC and the liquidity of the issuer. The results show that both counter-party risk and the risk of legal costs can significantly affect the value of ECs.

Apart from the articles listed above, there seem to be little research related directly to the valuation of ECs with an option structure. However, there are a multitude of research fields which are related to the problem of valuing ECs. For instance, there is a large body of research on the valuation of real options within the management literature on the topic of evaluating financial projects.

1.3 Purpose and Research Question

As described in the introduction, research into the valuation of ECs with an option structure is limited, for the estimation of the expected pay-out from EC contracts with an option structure.

The aim of this thesis is to evaluate how to best estimate the expected pay-out for ECs, testing various estimation methods against each other such as ARIMA time series models and GBM simulation. Furthermore, an attempt is made to adjust the models for bankruptcies and to incorporate implied market volatility to improve the estimation of the expected pay-outs.

Thus, the research questions of this thesis are:

- *How can the valuation of ECs be estimated using quantitative methods?*
- *Can adjustments for bankruptcy risks of the acquired company be incorporated to improve valuation accuracy?*
- *Can implied volatility from listed options be used to improve the valuation?*

2 Earnouts and Their Use

An EC is a contingent claim on a future pay-out given that certain objectives are met [Grønli and Hussein, 2021]. This is often that a specific financial outcome materializes. The result is that, instead of paying the seller of the business everything up-front, parts of the payment is deferred through an option contingent on the future performance of the business after the acquisition. ECs can take a multitude of different forms and structures. There are ECs that are binary in their structure, e.g., if a company receives market approval for their new drug after the acquisition the seller will receive a fix amount. Other ECs are linear in their pay-out structure, e.g., the seller will receive three times the company's profits during the first three years after the sale of the company. Other times ECs are structured similarly to an option, e.g., the seller of the company will receive a specified share of the profit in three years' time above a threshold K. The ECs with option structures are the most interesting from a valuation standpoint and therefore are the focus of this thesis [Chamberlain and Elkounovitch, 2019].

There are distinct reasons for using an EC during the acquisition of a company, several of which will be discussed in this thesis. However, the most obvious reason for using an EC is to get past differences in perceived value of the company [Grønli and Hussein, 2021]. It is often the case that the person looking to sell their company expects a high price and the actor looking to buy companies are looking for a lower valuation due to different uncertainties. The problem is that due to information asymmetry, the person selling the company will have better knowledge of the fair value of the company. Using ECs in these situations allows for an up-front price that the buyer is comfortable with, while giving the selling party a high potential pay-out if the company performs as expected [Choi, 2016]. Thus, ECs can be used to facilitate transactions and help the buyer and seller to agree on a price.

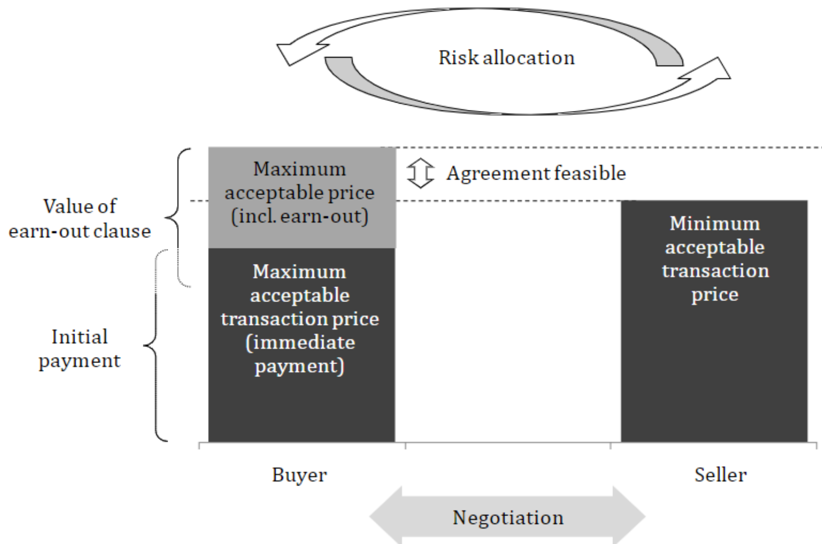

Figure 1: Illustration of how ECs manage the information asymmetry and differences in perceived value [Tallau, 2009]

2.1 The Rise of Earnout Contracts

During recent years there has been a strong consolidation trend where companies have merged or been acquired as global markets are becoming increasingly competitive [Flannery et al., 2020]. Being larger can lead to benefits such as economies of scale and higher probability of winning public procurement [Buchi et al., 2018]. This effect of consolidation and high mergers and acquisition (M&A) activity has been seen across industries such as life insurance, financial services, the hospital industry, and the airline network [Berger et al., 1999] [Dafny et al., 2014] [Dennis, 2005] [Cummins et al., 1999].

Within the world of M&A, it is becoming increasingly common to use ECs to facilitate transactions. Studies from the 1990's ECs showed that ECs were only used in around 3% of transactions [Datar et al., 2001] [Kohers and Ang, 2000]. Recent papers from 2018 shows that the use of ECs in transactions has now increased to encompass 15% of all acquisition transactions [Allee and Wangerin, 2018] [Bates et al., 2018]. It is thus clear that ECs are becoming more common as a means of facilitating acquisitions globally.

2.2 Reasons for Using Earnout Contracts

The main reason for using ECs has already been presented. In particular, the focus has been on how ECs can facilitate acquisitions by helping the buyer and seller agree on the price level. However, this is only one reason for why a company would potentially want to use an EC as a part of an acquisition. Within the research literature, the most common reasons for utilizing ECs within acquisitions have been found to be:

- **Agree to disagree:** When the buyer and seller cannot agree on a price. They can put the price difference as a contingent consideration. The buyer takes on less risk and the seller have the potential to receive a higher potential pay-out [Grønli and Hussein, 2021].

- **Manager retention:** An EC can be used for the purpose of retaining key managers in the company after the acquisition. Furthermore, it can function as an incentive to further motivate key managers [Barbopoulos et al., 2018].

- **Risk transfer:** When there is great uncertainty about the future value of the company. The buyer can transfer parts of the risk to the seller. As a result, acquisitions can more easily be made during times of uncertainty.

- **Increased leverage:** When a buyer puts parts of the transaction value in an EC, they are pushing parts of the payment into the future. This in practice is a way to finance the deal and increase leverage on the investment [Bates et al., 2018].

- **Honest signal:** If the seller is willing to accept an EC, then it is an honest signal of the manager's belief in the potential of the company. If a manager does not accept parts of the transaction value in the form of an EC, it could signal a lack of faith in the prospects [Reuer, 2005].

As indicated above, there are major benefits for using ECs during M&A transactions. These benefits are further indicated by the extensive literature showing that when a public company performs an acquisition with an EC, it is associated with a positive share price reaction. This indicates that investors view ECs as a positive component of an acquisition [Barbopoulos and Sudarsanam, 2012] [Barbopoulos et al., 2018] [Kohers and Ang, 2000] [Kohli and Mann, 2013].

2.3 Challenges With Using Earnouts

Despite the positive effects from using ECs in acquisitions presented above, there are potential pitfalls when using ECs. When utilized without enough foresight and experience, they can easily turn into more of a headache and financial liability than the benefits they promise. The most significant pitfalls exemplified in the research literature are given here below:

- **Integration Barriers:** When a company is acquired the purpose is to integrate it within the rest of the businesses of the buyer to achieve synergies. However, to measure the performance of the acquired company after the acquisition it is preferable to keep them as separate entities during the EC period [Bruner and Stiegler, 2008].

- **Destructive Incentives:** It is important to base the EC on the right metric because incorrect structures can lead to destructive incentives. For instance, an EC on the future sales growth could lead managers to pursue growth with little regard to the profitability of the business [Alexakis and Barbopoulos, 2020].

- **Legal Disputes:** ECs are the most common cause of legal disputes associated with M&A transactions. When taking potentially high legal cost from ECs into account, using ECs could yield lower returns [Battauz et al., 2021].

- **Future Burden:** Managers with a short-term perspective could view ECs as a source of liquidity to finance acquisitions when the buyer cannot afford the deal. Using ECs to fuel acquisitions today could turn into a drag on future company performance when payments are due.

- **Investor Protection:** ECs are less common in international mergers and acquisitions in countries with weak investor protection and problems in legal enforceability. Investors are thereby uneasy regarding delayed contingent payments when trust in legal enforcement is low [Reuer et al., 2004].

To summarize, there are clear benefits to ECs and investors tend to react positively when acquisitions include an EC. However, ECs are complex instruments that can cause problems down the line when they are not used properly.

2.4 When and How are Earnouts Used?

Given the positive and negative trade-offs related to using ECs presented earlier, it is of interest to see under what conditions they are used in practice.

ECs are used most often in acquisitions of private companies, in cross industry acquisitions where the acquirers are less experienced or during acquisitions of start-ups [Roncagliolo, 2020]. Furthermore, it has been shown that ECs are more common in tech-heavy or research driven biotech acquisitions. ECs are used less often when the buyer is much larger than the acquired company. A reason for this is that small companies might be hurt more for overpaying in transactions [Cain et al., 2011].

Moreover, it has been shown that ECs can be used to finance acquisitions for companies that are short on liquidity since part of the payment to perform the acquisition is delayed. It was found that companies with less liquidity acting in financing constrained markets are more likely to use ECs [Bates et al., 2018].

There are currently disagreements in the literature regarding how much of the deal value should be included in the EC. A rule of thumb is that the difference between what the buyer is willing to pay and what the seller is willing to accept should be placed within an EC. However, Lukas et al. [2012] considered an alternative approach of using game theory to estimate the optimal EC share of the total deal value. Showing that the simplified rule of thumb is not always optimal.

2.5 Earnout Accounting

Since ECs are difficult to value with high accuracy, they used to be accounted for when the EC started to reach maturity and it becomes clear whether the contract is likely to end up in the money or not [Cadman et al., 2014]. However, new rules and regulations (SFAS 141(R)) mandates continuous "fair valuation" of ECs to improve the accounting transparency of these instruments. This means that one should account an EC as a liability from when it is first issued. Furthermore, the regulations also mandate that ECs must be valued each financial quarter so that proper adjustments to the fair value can be made. In practice this means that an EC is accounted for as a liability and when reaching maturity, either in or out of the money, there will be a corresponding positive or negative effect on the reported earnings in the income statement. As a result, companies may attempt to modify their EC valuation in order achieve accounting objectives such as deferring tax payments [Ferguson et al., 2021].

2.6 Selection of Underlying Metric

Choosing a suitable underlying metric to base the option upon is an essential part of structuring an EC [Battauz et al., 2021]. Choosing the proper financial metric is essential to avoid creating destructive incentives and avoiding future legal disputes. According to Arzac [2004] an example of a good metric for an EC is to base it upon earnings before interest, taxes, depreciation, and amortization (EBITDA), adjusted for receivables older than one year. This measure is suitable because it does not discourage investment since depreciation's are not included. However, all businesses are different and what constitutes a suitable underlying metric for an EC will vary between each transaction.

During this thesis, simpler financial metrics will be considered as the underlying basis of ECs such as sales and EBIT. The reason is that there is more publicly available data with regards to sales and EBIT as they are metrics in line with generally accepted accounting practices (GAAP) [Insider, 2022]. However, it is noteworthy that during an actual transaction one would have access to EBITDA and other detailed financial figures.

3 Earnout Valuation Background

Valuing ECs is challenging and often not a straightforward task. This chapter introduces the valuation of ECs and covers the entire evaluation process. The goal is to give a broad introduction to EC valuation not limited to the estimation of the expected pay-out.

3.1 Valuation Principles

3.1.1 Probability Measures

There are two common probability measures that are used when valuing financial assets. The first perspective is risk neutral valuation under the probability measure $\mathbb{Q}$ and the second perspective is valuation under the measure $\mathbb{P}$ of actual probabilities [Meucci, 2011].

- $\mathbb{P}$: The measure of actual probabilities refers to the real probabilities for future outcomes. When working in $\mathbb{P}$ investors estimate the actual future expected returns and discount these for the risks and uncertainties associated with the future expected pay-out.
- $\mathbb{Q}$: The measure of risk neutral probabilities refers to a probability measure adjusted so that the probabilities incorporate the uncertainty of future returns. The expected return under $\mathbb{Q}$ is the risk-free rate and no discounting of additional risks is needed.

Using the probability measure $\mathbb{P}$ to estimate future expected real returns and discounting by a factor incorporating the riskiness of the returns is theoretically equivalent to valuation under the risk adjusted probability measure $\mathbb{Q}$.

3.1.2 $\mathbb{Q}$ Pricing vs $\mathbb{P}$ Pricing

As described above there are significant differences between working with the probability measure $\mathbb{Q}$ vs $\mathbb{P}$. What measure is desirable to work with depends on the application. In short it can be summarized that:

- Working in $\mathbb{Q}$ involves extrapolating the present using risk neutral probabilities for future outcomes found in the current market prices. The extrapolation under the probability measure $\mathbb{Q}$ is performed using continuous time martingales.
- Working in $\mathbb{P}$ often involves modelling future outcomes for risk management or portfolio management purposes. The quantitative modelling is often performed using multivariate statistics such as time series methods. The main challenge is to get high accuracy in the estimations.

[Meucci, 2011] Working in the probability measure $\mathbb{Q}$ has to a large extent dominated stock options pricing due to the Black Scholes formula. However, ECs are assets where neither the options nor the underlying metrics are publicly traded. Therefore, risk neutral pricing is not possible and thus relies on estimations of expected pay-out under the probability measure $\mathbb{P}$, and the estimation of discount rates for risk factors.

3.1.3 Discount Rate

When working in the probability measure $\mathbb{P}$, the challenge is to estimate the expected pay-out accurately by modelling future outcomes and discount the pay-out by the riskiness of the return at the appropriate levels. The discount factor incorporates a risk-free component referred to as the risk-free interest rate to compensate for the time value of money, and a risk premium to compensate for additional risks associated with the EC. There are several sources of risk premium for an EC, with the most important ones summarized below:

- **Non-diversifiable pay-out uncertainty**: The Capital Asset Pricing Model (CAPM) states that investors will require higher returns to take on risk that cannot be diversified [Chamberlain and Elkounovitch, 2019].
- **Counter party risk**: There is a probability that the counterpart defaults on payment when the EC ends up in the money Battauz et al. [2021].
- **Legal costs**: It has been argued that since ECs can lead to future legal conflict, investors should be mindful of this risk.
- **Liquidity premium**: It is well established that investors require higher returns to enter low liquidity financial positions. In the case of ECs, there is often no liquidity at all [De Jong and Driessen, 2006].

An in-depth discussion of the estimation of the non-diversifiable pay-out risk for an EC can be found in [Chamberlain and Elkounovitch, 2019]. As a high share of previous research has been focused on these factors, they have been left outside the scope of this report.

The counter party risk and legal risks for ECs have been studied and analysed by Battauz et al. [2021]. It was found that both risks can have a significant impact on the value of an EC when considered during modelling.

The liquidity risk premium has been estimated around 1.5% for speculative grade corporate bonds according to De Jong and Driessen [2006] while it has been found to be at around 3% for private equity [Franzoni et al., 2012].

3.2 Valuation Methods for ECs Without Option Features

3.2.1 Multiple-based ECs

A multiple-based EC is structured without option properties and is usually based on a financial metric for the company. An example of such a metric could be future revenue or profit. The EC pay-out is a linear function as it is a multiple of the underlying metric. An example is if an EC is constructed such that the seller will receive a multiple, α, times the earnings before interest and tax (EBIT) in T years after the acquisition. Thus, the ECs pay-out is a linear function of the profit in year T. The price of the EC can then be estimated using the expected value of the underlying metric, discounted by the discount rate r which incorporates the risk-free rate and the risk premiums. The price of the linear EC, Π_t, can thus be calculated as:

$$\Pi_t(\mathbf{X}) = \alpha\, \mathbb{E}^{\mathbb{P}}[\Pi_T(\mathbf{X})|\, \mathcal{F}_t]\, \mathrm{e}^{-r(T-t)}$$

[Chamberlain and Elkounovitch, 2019]. Here, α is a coefficient such that $\alpha \in \mathbb{R}^+$, $\mathcal{F}_t$ is the filtration, containing all known information about the underlying at time t, Π_t is henceforth used to represent the price or value of the input at time t

3.3 Earnout Contracts as Options

3.3.1 Relationship to Stock Options

Before considering ECs with an option structure, the valuation of common stock options is discussed to place the valuation of EC options in the context of option valuation. As previously discussed, stock options for publicly traded companies are generally valued under the probability measure $\mathbb{Q}$ with the discounted stock price as a continuous time martingale. In other words, the current price of the stock incorporates all available information regarding its future risks and returns. Thereby, the stock can be modelled as a GBM with the risk-free rate as drift. The price of a stock option can thus be described by a partial differential equation with the analytical solution being the Black Scholes formula.

Let S_t denote the price of a stock at time t, with $t = 0$ representing the present year. Let r refer to the risk-free rate, σ is the standard deviation of the stock returns. Let the price of a call option be referred to as $C(S_t, t)$, where K is the strike price of the option, T is the time of maturity for the option and $N(\cdot)$ denotes the standard normal cumulative distribution function. Then the Black Scholes formula is given by:

$$\begin{aligned}
C(S_t, t) &= N(d_1)S_t - N(d_2)K\, \mathrm{e}^{-r(T-t)} \\
d_1 &= \frac{1}{\sigma\sqrt{T-t}}[ln(\frac{S_t}{K}) + (r + \frac{\sigma^2}{2}(T-t)] \\
d_2 &= d_1 - \sigma\sqrt{T-t}
\end{aligned}$$

[Davis, 2010]. The model is based on assumptions that are discussed in more detail in Section 3.4. The only parameter that cannot be directly observed in the market is σ, the future standard deviation of S. As a result, the Black-Scholes formula is often used to calculate the implied volatility of an option given a certain price. Market participants will judge if the option is cheap or expensive given the implied volatility priced into the option and correct any arbitrage opportunities [Rutkowski, 2014].

The Black-Scholes model is used to a considerable extent in the market today, however the strict assumptions can provide some limitations to its use. Modified versions of Black-Scholes to generalize the conditions under which the formula can be successfully applied, such as to allow for dividends becomes increasingly common today [Davis, 2010].

3.3.2 Earnout Contracts as European Call Options

ECs often have pay-out functions that hold similarities to financial options, e.g., European call options. However, since neither ECs nor the underlying metric are publicly traded in a marketplace, the valuation methodology will differ from the valuation of stock options.

For an EC, the underlying metric of the option is not based on the stock price of a publicly traded stock. Instead, the underlying variable is a financial metric of a company that is often privately owned. Let $\mathbf{X}_t$ be the stochastic variable of the underlying metric of the option. Let t denote the time, K the strike price of the option, and T the options maturity. Then one can write the pay-out of the option at maturity as $V(\mathbf{X}_T)$, where $V : \mathbf{X} \mapsto \mathbb{R}^+$ and henceforth is used to denote the pay-out function of the EC. As an example, let us assume that the option is structured as a European call option on $\mathbf{X}$. Then the pay-out at maturity is:

$$V(\mathbf{X}_T) = max(\mathbf{X}_T - K, 0)$$

As discussed in Section 3.1.1, since ECs with option structures have underlying metrics that are not publicly traded in an arbitrage free market, they must be priced under the probability measure $\mathbb{P}$. The resulting price function $\Pi_t(\mathbf{X})$ of the EC thus depends on the expected pay-out of the EC under the probability measure $\mathbb{P}$ and the discount rate r, which includes the risk-free rate and risk premiums. The price of the EC can thus be written as:

$$\Pi_t(V(\mathbf{X}_T)) = \mathrm{e}^{-r(T-t)}\,\mathbb{E}^{\mathbb{P}}[V(\mathbf{X}_T)|\,\mathcal{F}_t] = \mathrm{e}^{-r(T-t)}\,\mathbb{E}^{\mathbb{P}}[max(\mathbf{X}_{\mathrm{T}} - K, 0)|\,\mathcal{F}_t]$$

3.3.3 Valuing Event Based Earnout Contracts

Event-based EC depends on the outcome of a future binary event. An example could be if the acquired company receives market approval from the FDA for a new drug. The valuation of a binary event, $\mathbf{X}_T \sim Bernoulli(p)$, depends on the probability p of the event occurring and the size of the potential pay-out C. Thereby, the pay-out, V, of the event-based EC can be written, $V(\mathbf{X}_T) = C\,\mathbf{X}_T$. The price of the event-based EC becomes the expected pay-out discounted by r. The main valuation exercise involved therefore relates to analysing previous outcomes of similar events to estimate the probability p and the discount rate r. The price of an event-based EC can thus be written as:

$$\begin{aligned}
&\Pi_t(V(\mathbf{X}_T)) = \mathrm{e}^{-r(T-t)}\,\mathbb{E}[V(\mathbf{X}_T)|\,\mathcal{F}_t] = \\
&= C\,\mathrm{e}^{-r(T-t)}(\underbrace{Pr(\mathbf{X}_T = 1)}_{p}\underbrace{\mathbb{E}[\mathbf{X}_T\,|\,\mathbf{X}_T = 1]}_{1} + Pr(\mathbf{X}_T = 0)\underbrace{\mathbb{E}[\mathbf{X}_T\,|\,\mathbf{X}_T = 0]}_{0}) = Cp\,\mathrm{e}^{-r(T-t)}
\end{aligned}$$

As these ECs are highly individual from case to case, this report only studies them briefly. A simple event-based structure is presented later.

3.4 Valuation Models

3.4.1 Black-Scholes for EC Valuation

The Black-Scholes model, as discussed above is a model used for the valuation of options. It is important to note that the Black-Scholes model relies on a multitude of assumptions which do not always hold true. The assumptions are as follows:

1. Constant volatility
2. Efficient market
3. No dividends
4. Interest rates constant and known
5. log-normally distributed returns
6. Only exercisable at maturity (European options)
7. No commissions and transaction costs
8. Perfectly liquid markets

[Janková, 2018]

For an EC, the most flagrant violations of the assumptions of the Black Scholes model are that the underlying metric of the option is not traded. As a result, there is no efficient market for the underlying metric and ECs are not perfectly liquid. Furthermore, since the underlying metric of an EC could theoretically be any metric related to the company e.g., when event-based, it is not always the case the returns of the underlying metric will follow a log-normal distribution. However, the assumption of log-normal returns seems to be somewhat valid for earnings before interest and taxes (EBIT) and sales, which are two of the most common underlying metrics for ECs [Rose, 2010]. Despite the violation of assumptions, there are researchers who apply the Black Scholes model to value options where the underlying metric is not traded. Options where the underlying metric is not traded is often referred to as *real options*.

To apply the Black Scholes model to ECs, attempts are made to "risk-neutralize" the underlying metric. This is done by using the generalized Black Scholes formula that allows for dividends. Then the growth and dividend yield are used to "risk-neutralize" $\mathbf{X}$ through incorporating the expected growth and the discount rate [Grant, 2011][Chamberlain and Elkounovitch, 2019][Tallau, 2009].

3.4.2 Monte Carlo Simulation for Earnout Valuation

Valuation by using Monte Carlo simulation is based upon sampling a distribution representing the underlying metric $\mathbf{X}_t$ of the EC. The repeated sampling of $\mathbf{X}_t$ results in a large number of different outcomes. When the sampling is large enough, the average of the expected pay-out converges to the true expected value because of the central limit theorem [Hoel and Krumscheid, 2019]. Monte Carlo simulation can thus be used as an alternative to the Black-Scholes model for valuing options. The distribution of $\mathbf{X}_t$ is often described as a GBM, supported by the fact that sales and EBIT returns show log-normal characteristics in previous studies [Rose, 2010].

Theoretically, when the Black Scholes model and Monte Carlo simulation is performed under the same assumptions, they should reach the same value for the EC. The main benefit of Monte Carlo simulation is the flexibility with which the distribution of $\mathbf{X}$ can be defined, while the main downside is the computational intensity needed for convergence. However, when valuing ECs, the simulation is done in discrete annual time increments, and the number of ECs to be valued is often low. Another benefit of Monte Carlo simulation is the possibility to value path dependent ECs such as American options with early exercise possible.

4 Background Theory

The purpose of this section is to outline the mathematical theory behind the models used in the thesis. Thus, giving a thorough background to the models used to generate the results of the thesis.

4.1 Introduction to Geometric Brownian Motion

4.1.1 The Wiener Process (Brownian Motion)

The botanist Robert Brown became in 1827 the first to introduce the concept of a Brownian motion [Malliaris, 1990]. Brownian motion refers to the random interactions and bouncing of particles suspended in a fluid [Klafter et al., 1996]. The Wiener process refers to the mathematical stochastic representation of Brownian motion developed by the mathematician Norbert Wiener [Masani, 2012]. The stochastic Wiener process, denoted as W_t with the increments $W_{t+u} - W_t$, is defined as:

- $W_0 = 0$
- $W_{t+u} - W_t$ is independent of W_s, for all $t > 0, u \geq 0, t \geq s$
- $W_{t+u} - W_t \sim N(0, u)$, where u is the variance
- W_t is continuous in t

The Wiener process has played a fundamental role within both applied and theoretical mathematics in the modelling of stochastic processes [Rutkowski, 2014].

4.1.2 Geometric Brownian Motion

GBM, or sometimes called exponential Brownian motion, is a continuous time stochastic process [Yor and Yor, 2001]. The logarithm of a GBM follows a Brownian motion with drift. A stochastic process in a financial application, $\mathbf{X}_t$, can be said to follow a GBM if it satisfies the following stochastic differential equation:

$$d\,\mathbf{X}_t = \mu\,\mathbf{X}_t\,dt + \sigma\,\mathbf{X}_t\,dW_t$$

with W_t being a Wiener process, μ as the expected drift and σ being the percentage volatility of the stochastic process [Marathe and Ryan, 2005]. The stochastic differential equation has the following solution using Ito's formula:

$$\mathbf{X}_{t+\Delta t} = \mathbf{X}_t^{(\mu - \frac{\sigma^2}{2})\Delta t + \sigma(W_{t+\Delta t} - W_t)}$$

Which can be represented as

$$\mathbf{X}_{t+\Delta t} = \mathbf{X}_t^{(\mu - \frac{\sigma^2}{2})\Delta t + Z\sigma\sqrt{\Delta t}}$$

where $Z \in N(0,1)$.

The GBM has most notably been used within financial modelling. The Black-Scholes formula, for instance, relies on modelling stock prices using the GBM [Marathe and Ryan, 2005].

4.1.3 Random Walk

One defining characteristic of the GBM is that it is continuous in time. However, for the simulation of ECs it is neither practical nor necessary to simulate continuous changes in the value of the underlying option metric. Instead, it is needed to simulate the changes from year to year. Thereby, when applied, the simulation will be a discrete random walk which is significantly less computationally intensive compared to simulating a continuous time process. It is noteworthy that if the step size is reduced for the random walk, it will eventually converge to a continuous GBM.

4.2 Time Series Analysis

4.2.1 ARIMA Model

To explain the Autoregressive Integrated Moving Average model (ARIMA) model, it is useful to first consider the Autoregressive Moving Average Model (ARMA) model. The ARMA model consists of one autoregressive part and one moving average part. The autoregressive part involves regressing the variable of the time series on its past lagged values. The moving average component of the ARMA model consists of modelling the error term as a linear combination of current error and error terms in the past. Mathematically an $AR(p)$ model, which refers to an autoregressive model of order p can be described as follows:

$$\mathbf{X}_t = c + \sum_{i=1}^{p} \delta_i \, \mathbf{X}_{t-i} + \epsilon_i$$

where $\mathbf{X}_t$ is the variables of the time series, which in the context of this thesis is the same as the underlying variable of the options, c is a constant, ϵ is white noise, p is the order of the AR model and δ represents to coefficients of the AR model in relation to the lagged instances of the variable [Ho and Xie, 1998].

The $MA(q)$ part of the model on the other hand refers to a moving average model of order q, which describes the error term as a linear combination of past errors. Mathematically this part of the model can be described as follows:

$$\mathbf{X}_t = k + \epsilon_t + \sum_{i=1}^{q} \theta_i \epsilon_{t-i}$$

In this instance, $k = \mathbb{E}[\mathbf{X}_t]$, q refers to the order of the MA model, θ refers to the parameters of the MA model and ϵ refers to white noise error terms.

As stated initially the ARMA model is a combined AR and MA model. Thus, one can write the $ARMA(p,q)$ model with the following mathematical representation:

$$\mathbf{X}_t = c + k + \epsilon_t + \sum_{i=1}^{p} \delta_i X_{t-i} + \sum_{i=1}^{q} \theta_i \epsilon_{t-i}$$

The ARMA model can be applied under conditions when the time series is stationary in both the sense of being mean stationary and the variance. Under circumstances where the time series is not stationary, due to it containing a growth trend, an extension of the ARMA model can be used, the ARIMA(p, d, q). The ARIMA model is similar in nature to the ARMA model, however it includes differencing of the time series, to make the time series mean stationary [Ho and Xie, 1998]. Using the lag operator, L, defined such that $L^k \mathbf{X}_t = \mathbf{X}_{t-k}$, the ARIMA model can be represented mathematically as follows:

$$(1 - \sum_{i=1}^{p} \delta_i L^i)(1 - L)^d \mathbf{X}_t = (1 + \sum_{i=1}^{q} \theta_i L^i)\epsilon_t$$

The ARIMA process utilizes the lag operator for differencing which is a systematic process for making non-stationary time series stationary by removing trend or seasonality. The difference operator ∇ is defined in terms of the lag operator as $\nabla^i \mathbf{X}_t = (1 - L)^i \mathbf{X}_t$. In short, the ARIMA process is an ARMA process where differencing is used to make the process mean stationary [Fattah et al., 2018].

4.2.2 ARIMA Prediction Interval

The closed form mathematical formula for the prediction interval (PI) volatility $\hat{\sigma}^2$ does not always follow a simple general mathematical pattern. However, for ARMA models a simple closed form expression can be given as follows.

$$\hat{\sigma}^2 = \sigma^2 \sum_{j=0}^{h-1} \hat{\psi}_j^{-2}$$

Where σ is the historically estimated residual variance and $\hat{\psi}$ refers to the coefficients for the moving average representation of the ARMA(p, q) process. Under normally distributed residuals, the PI of the prediction becomes:

$$\hat{x}_T \pm z_{\alpha/2} \hat{\sigma}(h)$$

The multi-period ARIMA time series forecast model utilized differencing and does not have a simple general closed form formula such as for the ARIMA(p,q) model. However, there exists models in R utilizing methods such as Kalman filtering to estimate the PI for each time series point estimate.

4.2.3 Prediction of Short Time Series

Time series analysis is most effectively utilized when there is widely available high frequency data. However, in the case of financial metrics for private companies, data is only available on an annual basis and the time series in this article can is of length 12-14 data points.

The widely cited book by Hyndman and Athanasopoulos [2018] writes that thirty data points for ARIMA models is often stated as a lower bound for data necessary to perform an ARIMA model without much evidence to back this claim up. Hyndman and Athanasopoulos [2018] argues instead that there is no lower boundary, and that the success of the implementation will instead depend on the broader context.

They suggest that how much data is needed depends on two components:

1. The amount of randomness in the data

2. The number of parameters to be estimated

If the randomness in the data is high, then more data is needed for the model to identify the patterns. Furthermore, there must be more datapoints than the number of parameters to be estimated.

A common evaluation measure for time series analysis is the Akaike information criterion (AIC), which is a proxy for the one-step out of sample performance of the forecast [Sakamoto et al., 1986]. AICc is a modification for short time series which includes a punishment for the number of parameters to be estimated [Brewer et al., 2016]. When applying the autoARIMA function in R to time series of length shorter than twenty data points, it is common for the function to fit models with between one to three parameters based on AICc score. However, in this thesis the judgement has been made to test model performance based on the actual out of sample forecasting error rather than AIC. This allows for testing of a larger set of models and reduces the risk of overfitting the models to the data [Hyndman and Athanasopoulos, 2018].

4.3 Artificial Neural Networks

Artificial neural networks (ANN) or neural networks (NN) is a class of machine learning algorithms that is inspired from how animal brains are constructed [Jain et al., 1996].

4.3.1 Artificial Neurons

The artificial neuron is the most fundamental component of an ANN. It was initially theorized by Rosenblatt's perceptron. Each neuron consists of an input vector $\mathbf{X} = (X_1, ..., X_n)^T$ and yields an output Y [Izenman, 2009]. Each X_i is linked to Y through a corresponding weight coefficient, w_i. Each neuron contains a transfer function (activation function), Φ, that is used to compute the output. The relationship between the input and output can be written as $Y = \Phi(\sum_{j=0}^{n} w_j X_j)$. To give an example, if $\Phi(X_i) = X_i$, it is equivalent to linear regression with coefficients w_i. The transfer function Φ can be selected as a multitude of different functions depending on the applications and can thus yield a large range of outcomes depending on the application [Jain et al., 1996].

4.3.2 Kolmogorov-Arnold Representation Theorem

Theorem 1 (Kolmogorov-Arnold Representation Theorem) *Let f be any continuous function such that $f : [0,1]^n \longrightarrow \mathbb{R}$. It then exists univariate continuous functions $g_q, \Psi_{p,q}$ such that*

$$f(x_1...x_n) = \sum_{q=0}^{2d} g_q(\sum_{p=1}^{d} \Psi_{p,q}(x_p))$$

[Schimdt-Hieber, 2020].

The theorem implies that it is theoretically possible to approximate any continuous function, given sufficient neurons and at least one hidden layer in the model and is fundamental to the use of ANN models.

4.3.3 Supervised vs Unsupervised learning

Supervised machine learning is a subset of machine learning algorithms, defined by the characteristic of using labelled data to train the algorithm [Sup]. The algorithms have a loss function that is minimized as training progresses. It is referred to as supervised learning due to the requirement of human input to label the dataset before training can start. The algorithms are often used for classification and regression problems.

Meanwhile, unsupervised learning is defined as machine learning algorithms which work directly on a raw, unlabelled dataset. These algorithms are often useful to identify patterns and solve clustering and association problems.

The ANN used in this article is part of the supervised learning category and is deemed as suitable due to the problem focusing on forecasting based on historical data.

4.3.4 ANN Architecture

On a high level, ANN models are composed by three different parts:

1. Input layer
2. Hidden layer
3. Output layer

[Jain et al., 1996]

Each layer consists of neurons, whose role is to receive information, process it and produces an output through the transfer function. The neurons in each layer have no mean to communicate with each other but passes on information to the neurons in the next layer. Thereby, there are several different ways in which an ANN-model can be constructed. First and foremost, there is the possibility to vary the number of neurons in each layer, which varies depending on the nature of the problem. Secondly, there is a possibility to vary the number of layers for the model. It is recommended that if the data is non-linear, at least one hidden layer should be used [Sarle, 1994].

Hyper-parameter tuning is critical to achieve the best possible predictive model. As there rarely is a set rule for how to set the hyper-parameters, a common way is to vary them independently of each other to find approximately optimal values using methods such as grid search.

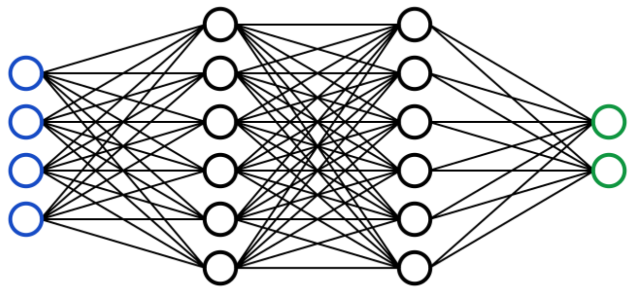

Figure 2: Three-layer NN

In this thesis, feed forward NN are used, which signifies that the information in the model only can travel forward, while in a recurrent NN model, the information can return to the layer and reiterate [IBM, 2020]. After testing several different combinations, performance measures showed that a model consisting of a single input, hidden and output layer performed the best results, and henceforth it is the type of model that will be referred to when talking about ANNs and related models. The model is further specified in the methodology section.

4.4 Hybrid Model

The hybrid model used in this thesis is a combination of ARIMA-models and artificial neural networks for forecasting time series based on work by Khashei and Bijari [2011]. The idea is to utilize the best aspects from both models, with the ARIMA model finding the linear relationships of the time series, while the ANN attempts to identify non-linear relationships in the residual to further improve the results [Babu and Reddy, 2014]. Several studies have in the past shown the model to yield better predictive results compared to the separate models. This is especially true for time series with a high prevalence of heteroscedasticity. Babu and Reddy [2014] showed that a hybrid model approach showed superior results when it comes to single- and multistep predictions of sunspots. Furthermore, Khusuwan and Waiyamai [2019] has shown that the model is effective at predicting *earnings before interest taxes depreciations and amortisations* (EBITDA) in retail companies. The model is constructed through first applying the ARIMA model to the time series data, and thereby extracts the underlying linear trend of the data. Thereafter, an ANN is constructed based upon the residual data to extract non-linear relationships from the data [Khashei and Bijari, 2011]. As in this case, there is limited information available due to the short time-series, it is theorized that the model should be more accurate than the other models as it can extract more information from the data.

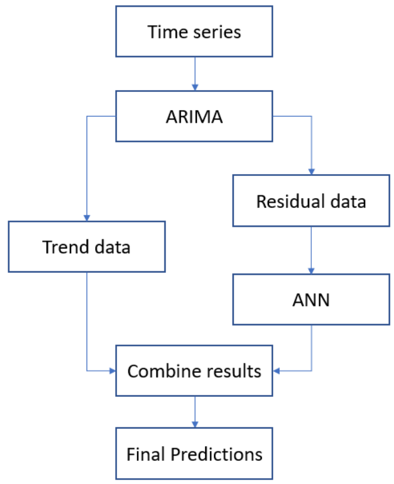

Figure 3: Flowchart over construction of hybrid model

4.5 Logistic Regression and Bankruptcy

4.5.1 Logistic Regression

The logistic regression model is a statistical model that estimates the probability of a binary outcome by having the logarithm of the odds be a linear combination of one, or more, independent predictors [Menard, 2002]. In logistic regression, the dependent variable is binary, and the two possible outcomes are coded as 1 and 0. The independent regressors can either be categorical variables or continuous variables. The function that converts the log odds into the probability of the event is the logistic function $p(x)$. The logistic function is a type of sigmoid function, and other binary classifiers can have a different sigmoid function than the logistic function.

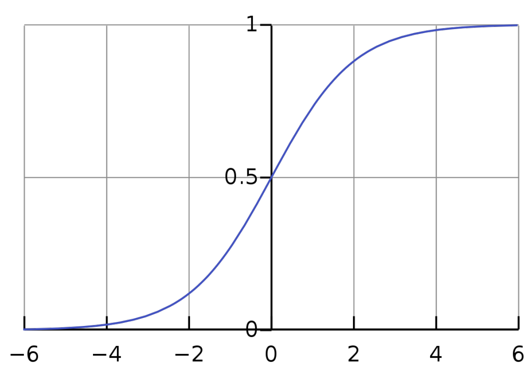

Figure 4: The standard logistic function

The logistic function is illustrated in the image above and can be written as follows:

$$p(x) = \frac{1}{1 + e^{-(x-\mu)/s}}$$

In this context μ refers to the location parameter or the value of x, such that the probability is $\frac{1}{2}$. The constant s on the other hand is known as the scale parameter. The function can be re-written on the following form.

$$p(x) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}}$$

[LaValley, 2008].

4.5.2 Previous Results on Predicting Bankruptcies

Predicting bankruptcies is an area of research that has been extensively investigated. Logistic regression is one of many methods that have been used for this purpose. The main challenge when predicting bankruptcy using logistic regression is that the datasets often are unbalanced. The average number of companies that goes bankrupt each year is around 1%, ranging between $0.5 - 2.2\%$ per year [Hillegeist et al., 2004]. Thereby, when taking a random sample of companies, the share of companies that have not gone bankrupt will be significantly higher. Logistic regression utilizes maximum likelihood estimation of regression coefficients, a process which is not suitable when the dataset is highly unbalanced[Bellovary et al., 2007].

The field of research has seen high prediction accuracy for research models [Bellovary et al., 2007]. However, it is important to note that prediction accuracy is problematic when evaluating the model on a highly unbalanced dataset. If the dataset is unbalanced, then a high prediction accuracy can be achieved by only guessing that the company will not go into bankruptcy. Thereby, an alternative measure is using weighted accuracy, and thereby reward a model for more accurately predicting companies that go bankrupt. This is done by studying each model's confusion matrix.

4.5.3 Outlier Analysis Logistic Regression

Let the residual be denoted as

$$\epsilon_i = y_i - \hat{y_i}, i = 1, 2, ..., n$$

where y_i is an observed value and $\hat{y_i}$ is the value given by the model [Montgomery et al., 2012]. To achieve a good model, the residual must be analysed to correct any abnormalities in the model. There are several approaches that can be used to analyse the residual, each having their advantages and disadvantages. A commonly used measure for residual analysis is Cook's Distance, which is used to identify influence points that strongly impacts the model. Cook's Distance is defined as

$$D_i = \frac{r_i^2}{p}\frac{h_{ii}}{(1-h_{ii})}, \ \ i = 1, 2...n$$

with r_i^2 being the squared studentized residual, and $\underline{h_{ii}}$ is a component of the hat-matrix [Montgomery et al., 2012]. A point i is classified as an influence point if $D_i > \beta$, where β is some cut-off value. There are some debates to which cut-off value to use. Cook proposed $F_{0.50}(p, n-p)$ to be a suitable cut-off level, where p and $n-p$ degrees of freedom and F is the $F-distribution$ [Cook, 1977]. This is the cut-off level being used in this report.

4.6 Implied Volatility

For options valuation the future volatility is essential since it has a significant impact on the value of options [Ball and Roma, 1994]. A common problem is that future volatility can differ significantly from historical volatility. As a result, the implied volatility from the current prices of options with expiration date in the future can be used as proxies for future volatility [Poon and Granger, 2005]. The implied volatility is in this case solved for through the Black-Scholes formula and is described in 3.4. Calculating the implied volatility of an option requires inverting the option pricing function to solve for volatility. However, there is no full closed form solution for the implied volatility[Chance, 1996]. As a result, one can either use closed form approximations e.g., Brenner Subrahmanyam approximation or the Li formula . Another approach for estimating the implied volatility from the Black Scholes formula is using methods for finding the numerical roots through e.g., Newton-Raphson [Orlando and Taglialatela, 2017].

5 Methodology

The focus in this thesis is on improving the estimation of the expected pay-out from ECs. The approach used was based on Monte Carlo simulation as it allowed for modifications of the distribution of $\mathbf{X}$, and for simulation of path dependent option structures.

Four different estimation methods were tested and compared against each other in terms of mean squared error of the actual pay-out. The methods used for modelling $\mathbf{X}$ were GBM, ARIMA, ANN and a hybrid model based on ANN and ARIMA.

To further improve the estimation, two modifications of the models were tested. Firstly, an attempt was made to adjust the distributions of $\mathbf{X}_t$ for the risk that the acquired company might go into bankruptcy. Secondly, an approach for adjusting the volatility of the models through incorporating the implied market volatility was tested.

5.1 Relationship Between Earnout Price and Expected Pay-out

Let V be the pay-out function for some earnout contract written on $\mathbf{X}$ with maturity T. Assume that the discount rate r is known. Then it holds that:

$$\Pi_t(V(\mathbf{X}_T)) = \mathrm{e}^{-r(T-t)}\,\mathbb{E}[V(\mathbf{X}_T)|\,\mathcal{F}_t] \longrightarrow \Pi_t(V(\mathbf{X}_T)) \propto \mathbb{E}[V(\mathbf{X}_T)]$$

.

As the models that are evaluated in this study did not influence the discount rate, all payments occurred at maturity and several previous studies have looked at how to set the discount rate when valuing ECs, each model's performance was evaluated in terms of expected pay-out at maturity. The results thereby focus on expected pay-out, and can easily be adjusted to get the price and value of the EC.

5.2 Estimating Expected Pay-out for ECs

Four different methods for estimating the expected pay-out of ECs were tested. The models were applied to 14 years of historical data for a large number of companies and were evaluated in terms of their one or two years out of sample prediction of the expected pay-out. The expected pay-out of each EC was determined either analytically or through Monte Carlo simulation. The four models were as follows:

- A GBM model
- An ARIMA time series model
- An ANN model
- An ARIMA / ANN hybrid model

Below each model is described in more detail. Covering how parameters were estimated, how the forecasts were performed, and how the expected pay-outs were computed.

5.2.1 Geometric Brownian Motion Model

The first approach was to model $\mathbf{X}$ as a Geometric Brownian Motion (GBM), which can be represented as:

$$\mathbf{X}_{t+\Delta} = \mathbf{X}_t^{(\mu - \frac{\sigma^2}{2})\Delta t + Z\sigma\sqrt{\Delta t}}$$

Where $\mathbf{X}$ refers to the underlying metric, μ is the drift, σ is the standard deviation of returns and $Z\ N(0,1)$. See 4.1.2 for a complete description of a GBM and the origin of the above representation. According to Hull [2003], the estimations for the parameters μ and σ were computed as follows:

$$u_i = \ln\left(\frac{\mathbf{X}_i}{\mathbf{X}_{i-1}}\right),\ i = 1, 2, .., n$$

with u_i representing the log returns compared to the previous year. The standard deviation $\hat{\sigma}$ was given by

$$\hat{\sigma} = \sqrt{\frac{1}{1-n}\sum_{i=1}^{n}(u_i - \bar{u})^2}$$

where $\bar{u}$ is the average log return.

Once $\hat{\sigma}$ was estimated, the value was used to estimate $\hat{\mu}$, given as follows:

$$\hat{\mu} = \frac{\ln(\mathbf{X}_t) - \ln(\mathbf{X}_0)}{t} + \frac{\hat{\sigma}^2}{2}$$

when the parameters $\hat{\mu}$ and $\hat{\sigma}$ had been estimated, t signifies the last available year, Monte Carlo simulation was used to generate simulations for the underlying EC metric by sampling Z from the standard normal distribution. The GBM was simulated according to the previously stated representation. For each simulation, the pay-out of the EC structure was computed. The average pay-out from all simulation for a company resulted in the expected pay-out.

5.2.2 ARIMA Model

The second valuation model was based on the ARIMA time series. A theoretical description of an ARIMA(p, d, q) model can be found in 4.2.1. ARIMA models were constructed on 12-13 years of data. As $\mathbf{X}$ is log-normally distributed, a log-transform was performed before running the model to ensure a linear relationship in the data. The transform was done due to suggestions from previous studies and observations of the time series that indicated that a log-transform was suitable. When calculating the expected pay-out, the model's prediction was re-transformed to get the value of the underlying metrics. Note that this was also the case for the ANN and hybrid models, but not the GBM as it already assumes log-normally distributed data.

After the coefficients were calculated, the ARIMA models were used to generate point forecasts for the expected outcome of the underlying metric $\mathbf{X}$ until maturity T. In this case the time until maturity was either one or two years depending on the pay-out structure that was estimated. Each point estimate of the forecast in turn yielded a prediction interval (PI), given as:

$$\hat{\mathbf{X}}_T \pm e^{\hat{\sigma} z_{\alpha/2}}$$

Where $\hat{\mathbf{X}}_t$ refers to the point forecast at time t, $\hat{\sigma}^2$ refers to the volatility of the standard error and $z_{\alpha/2}$ refers to the two-tail percentage point of a standard normal distribution at level α [Chatfield, 2001].

Multi-period PIs for ARIMA models also follow a normal distribution if the residuals of the model are normally distributed. However, the formula for the variance of the prediction error does not have a simple general solution [Hyndman and Athanasopoulos, 2018]. In this thesis the ARIMA package in R was used which contains methods for calculating the multi-year PI.

A limitation of the PI is that it only takes the error arising from the model residuals into account. Thereby, there were several implicit assumptions made with regards to other potential sources of errors:

- It was assumed that an ARIMA model is the correct model to represent the data.
- It was assumed that the model coefficients are estimated accurately without error.
- It was assumed that historical data is entirely representative of future outcomes

[Chatfield, 2001]. Under these assumptions, the ARIMA forecast point estimate together with the volatility of the PI could be used compute the expected pay-out of the EC through Monte Carlo simulation or analytical solutions. A potential risk was that the PIs can tend to be too narrow, thereby limiting the prediction accuracy when looking at expected pay-out. A detailed pseudo-code of the algorithm for computing the expected pay-out can be found in 11.1.

5.2.3 Artificial Neural Network

An Artificial Neural Network (ANN) based regression model was used, implemented through the "nnetar: Neural Network Time Series Forecast" package in R. The package is used to forecast univariate time series. The model used was a feed-forward neural networks with a single hidden layer and lagged inputs and was implemented with supervised learning. More background theory on ANN methods can be found in 4.3.

The model training data used in the input layer was 12 years of time-series data for the underlying metric (sales or EBIT). The test data consisted of the latest out of sample value of the time series. Furthermore, the ANN was constructed with an architecture in the form of one input layer, one hidden layer and one output layer with respectively 13/7/1 nodes in each layer. Grid search was utilized to determine the optimal number of layers and nodes. The model returns a forecasted value and the corresponding volatility of the forecast. As an ANN is can be seen as a function approximation, it is however not as intuitive what the volatility represents or how it should be computed. In this report, empirical PIs based on Pearce et al. [2018] methodology of High-Quality PIs for Deep learning that uses parameter and bootstrap resampling. An in-depth description of the methodology can be found in the original paper. In general, these PIs are distribution less, but were assumed to follow a normal distribution. Thereby, an estimate of the model's volatility could be obtained. The assumption of normally distributed PIs was strengthened by studying the QQ-plot.

5.2.4 The Hybrid Valuation Model

The final method for estimating the expected EC pay-out was to use a hybrid model based on ARIMA and ANN. The estimation process for the hybrid model was equivalent to the ARIMA and ANN methods. The main difference was that the ANN used the residuals of the ARIMA model as training data instead of the initial time series. The R package NNETAR was used to create the model. The model returns both the forecast point estimate and corresponding volatility. These values were then used to Monte Carlo simulate the expected pay-out for each EC and compare it to the actual pay-out given the out of sample data. See 4.4 for further details on the construction of a Hybrid model.

5.3 Model Adjustments to Improve Accuracy

The ARIMA, ANN and Hybrid models previously described were implemented under multiple simplifying assumptions. To increase the accuracy of the models, an attempt was made to generalize the model by incorporating the risk of bankruptcies and improved estimation of volatility through implied volatility of listed peers.

- **Bankruptcy:** The models assumed that the historical data is representative for future outcomes. However, the historical data did not contain outlier outcomes such as bankruptcy. Adjusting for bankruptcy outcomes could thereby improve the prediction accuracy.
- **Market volatility:** The time series models assumed that the PI volatility is a good estimator of the future volatility prediction. However, previous research had shown that there is a tendency for this to be underestimated. Therefore, volatility estimates were attempted to be improved through incorporating parts of the implied future market volatility.

5.3.1 Bankruptcy Adjustment of Underlying Metric Distribution

In the three methods above, for estimating the expected pay-out, it was assumed that historical data was representative of future outcomes. However, it was known that there were outcomes that had not been represented in the historical data, which could have a high impact on the EC valuation. The most prevalent of these was if the acquired company went into bankruptcy due to unforeseen aspects such as fraud or an accident. This type of events is not represented in historical data, as the acquirer cannot purchase a bankrupt company. Thereby, if not incorporating the bankruptcy risk in the valuation there was a risk of survivorship bias in the estimation of the expected pay-out from the EC [Brown et al., 1992].

A hypothesis was presented that the models could be improved by adjusting the expected pay-out by bankruptcy risk, where the value of the EC thereby should be zero. This bankruptcy adjustment was inspired by a previous study by Stein and Jordão [2005], that used bankruptcy adjustment to calculate income volatility of companies more accurately. The bankruptcy adjustment for the time series processes was performed by defining a new stochastic variable $\mathbf{X}_t^{(b)}$, which was conditional on if the company has gone bankrupt or not. Let Y be a stochastic variable representing the year in which the company goes bankrupt and d be the probability that the acquired company goes bankrupt in the following year, computed through logistic regression, and assumed to be constant over the forecasted period. Then

$$Y \sim geometric(d) \longrightarrow p_Y(y) = d(1-d)^{y-1}$$

and thereby the new variable was modelled as

$$\mathbf{X}_t^{(b)} = \begin{cases} \mathbf{X}_t & t < Y \\ 0 & t \geq Y \end{cases}$$

Where $\mathbf{X}_t$ is the same stochastic variable that has been used previously in the report. Thereby, the expected pay-out of the EC under the assumption of Bankruptcy risk could be computed as follows:

$$\begin{aligned} \mathbb{E}[V(\mathbf{X}_T^{(b)})] &= Pr(Y > T)\,\mathbb{E}[V(\mathbf{X}_T^{(b)})|Y > T] + Pr(Y \leq T)\,\overbrace{\mathbb{E}[V(\mathbf{X}_T^{(b)})|Y \leq T]}^{0} = \\ &= \underbrace{Pr(Y > T)}_{(1-d)^{T+1}}\,\underbrace{\mathbb{E}[V(\mathbf{X}_T^{(b)})|Y > T]}_{\mathbb{E}[V(\mathbf{X}_T)]} = (1-d)^{T+1}\,\mathbb{E}[V(\mathbf{X}_T)] \end{aligned}$$

Thus, when the expected pay-out was calculated, most of the pay-outs were sampled as in the previous models. However, parts of the time the company was assumed to have gone bankrupt, yielding a pay-out of zero. The bankruptcy risk for each acquired company, d, was estimated by constructing a logistic regression model on a large dataset of Swedish companies with eleven tested regressors. The investigated regressors were:

- Revenue
- EBIT
- Long-term Debt
- Cash and cash equivalent
- Share Holder Equity
- Revenue 3-year CAGR
- EBIT-Margin
- Cash and cash equivalents / Revenue
- Share Holder Equity / Revenue
- Long term debt / Shareholder equity
- Long term debt / Revenue

5.3.2 Estimating Volatility

The use of implied volatility was implemented through a linear combination of the models own predicted volatility and the implied volatility of listed companies. A volatility estimate based upon the Euro Stoxx 50 Volatility (VSTOXX) index was used to represent the markets implied volatility for Europe. The index is based upon implied volatility estimates from options on fifty large blue-chip companies in the European area [Quontigo, 2022]. The hypothesis was that a part of the acquired company's volatility was influenced by the market volatility, which could be measured through the STOXX50. This hypothesis was supported by previous works from Cipollini and Gallo [2019] which argued that a shared market volatility component within European markets represented by the STOXX 50 volatility exists.

To choose a representative volatility, the average implied one-year volatility over the last three months of the training data calendar year was used. Let σ^2_{VSTOXX} denote the average implied volatility squared, and σ^2_{model} denote the volatility suggested by the model. New estimates of the underlying metric volatility σ^2_{tot} was composed as a combination of σ^2_{VSTOXX} and σ^2_{model} through:

$$\sigma^2_{tot} = \alpha\sigma^2_{VSTOXX} + (1-\alpha)\sigma^2_{model}$$

where $\alpha \in \{0, 0.1 \ldots 0.9, 1\}$. Each model was then evaluated using varying values on α to determine if there were any advantages in incorporating a market volatility component.

5.4 Simulation of Expected EC Pay-outs

5.4.1 Contract Structure

The ability of the models to predict expected pay-outs was tested on a prediction ability against real outcomes. This was done for three different EC contract structures. Let K denote the strike price and $\mathbb{1}$ denote the indicator function. Then the pay-out functions were as follows:

Binary Option Pay-out: The binary option was an option with two years to maturity and with the strike price set as $K = \mathbf{X}^{(i)}_{T-2}$, where T is the last year in the time series. This was then equivalent to the value of the underlying metric when the EC was issued. The pay-out structure was evaluated for the time series for all companies, and indexed through $i \in \{1, 2, ..., n\}$ in the dataset. The structure could then be expressed as:

$$V(\mathbf{X}^{(i)}_T) = \mathbb{1}_{\{\mathbf{X}^{(i)}_T > \mathbf{X}^{(i)}_{T-2}\}}$$

Linear Option Pay-out: A linear option pay-out was tested with one year to maturity and strike price set to $K = 1.05\,\mathbf{X}^{(i)}_{T-1}$. The option structure could then be expressed as:

$$V(\mathbf{X}^{(i)}_T, \mathbf{X}^{(i)}_{T-1}) = (\mathbf{X}^{(i)}_T - 1.05\,\mathbf{X}^{(i)}_{T-1})\mathbb{1}_{\{\mathbf{X}^{(i)}_T > 1.05\,\mathbf{X}^{(i)}_{T-1}\}}$$

Path Dependent Pay-out: For the path dependent pay-out EC, the pay-out did not only depend on $\mathbf{X}_T$, but also the previous values affected the pay-out. In this instance, the time to maturity was $T = 2$ years and the strike price was the lagged value $\mathbf{X}_{t-1}$. The structure could then be expressed as:

$$V(\mathbf{X}_T^{(i)}, \mathbf{X}_{T-1}^{(i)}) = max(\mathbf{X}_T^{(i)} - \mathbf{X}_{T-1}^{(i)}, \mathbf{X}_{T-1}^{(i)} - \mathbf{X}_{T-2}^{(i)}, 0)$$

5.5 Other Implementation Aspects

5.5.1 Outlier Identification in Time Series

To identify time series that contains potential outliers, Generalized Extreme Studentized Deviate (Generalized ESD) was used. Generalized ESD is performed through first comparing the residual to the rest of the time series [Montgomery and Vining, 2012]. Let $\{\mathbf{X}_t\}_{t=1}^{N}$ be a time series of some stochastic process containing N datapoints. Let $\hat{x}_t$ be a prediction from a model at time t, ϵ_t be a residual of a model defined as $\epsilon_t = x_t - \hat{x}_t$, where x_t is the actual value and $\hat{x}_t$ is the value predicted by the model, $\overline{x}$ is the mean value of the time series, R_i is a test statistic. Then for each time series, a hypothesis test can be formulated as follows:
H_0 : The dataset does not contain more than r outliers
H_1: The dataset does contain more than r outliers.

The procedure to test the time series for outliers is as follows:

1. $\overline{x} = \sum_{\forall i \in S} \frac{x_i}{N}, S = \{1, ..., N\}$
2. Compute $\epsilon_i = x_i - \overline{x},$ for all $i \in S$
3. $\sigma^2 = \sum_{\forall i \in S} \frac{(x_i - \overline{x})^2}{N}$
4. Compute the test statistics $R_i = \frac{max|x_i - \overline{x}|}{\sigma^2},$ for all $i \in S$
5. $T_i = \frac{(T-i) \cdot t_{\alpha, T-t-1}}{\sqrt{(T-t-1+t^2_{\alpha,T-i-1}) \cdot (T-t+1)}}$
6. $Z_i = \begin{cases} 1 & \text{if } \mathrm{R}_i > \mathrm{T}_i \\ 0 & \text{otherwise} \end{cases}$
7. If $\sum_{i \in S} Z_i > r$ reject H_0

where S is the set of indices of the residual, N is the number of elements in S, $t^2_{\alpha,T-i-1}$ is t-distributed with $T - i + 1$ degrees of freedom at confidence level $\frac{1-\alpha}{2(T-i+1)}$ and significance level α.

5.5.2 Monte Carlo Convergence

For Monte Carlo estimates to be accurate, it was required to run enough simulations for the algorithm to converge. Meanwhile, running too many simulations would in turn make the computations significantly slower with low additional benefit. To reach high computational efficiency, the *acceptable shifting convergence band rule* (ASCBR), developed by M.Y.Ata [2007] was used. A detailed theoretical derivation of the rule is to be found in the original work. In short, the Monte Carlo algorithm kept running until the estimate of the expected pay-out was contained within a specific interval width for ζ consequent runs. Based on a previous study by Ciaccia et al. [2016], $\zeta = 100$ was deemed a good fit for this type of problem.

6 Data

The following section is dedicated to exploring the datasets used in the thesis. These are later used to estimate parameters and to build the valuation models.

6.1 Data Source

The main source of data for all the models was provided through the database Orbis, which gives access to financial data of private companies and is of particularly high quality for Nordic companies. Orbis has become a common source of data for researchers interested in studying private companies that are not publicly traded. Several studies have investigated the quality of the data and how representative it is, and they have found the database reliable with some minor shortcomings [Bajgar et al., 2020] [Ribeiro et al., 2010] [Kalemli-Ozcan et al., 2015]. Orbis data was therefore used as a foundation for the time series analysis and the logistic regression for bankruptcy prediction.

6.2 Dataset 1: Earnout Pricing

6.2.1 Description

The first set of models constructed for this thesis consist of various time series models to forecast the underlying metric of the EC. The data used for the time series forecasting is annual EBIT and sales data of length fourteen for a set of private Nordic companies with a minimum of $2m in annual revenue. The focus on the Nordics is due to the higher prevalence of financial data, due to regulations on financial transparency [Aisbitt, 2001].

6.2.2 Search Strategy

The search conditions that were used to generate the dataset were as follows:

- Status: Active companies
- Country/Region: Sweden
- Operating Revenue: Minimum 2 million USD
- Standardized Legal Form: Private limited company
- Entity type: Corporate

The search resulted in an initial dataset of 21 137 companies before removing outliers and missing values.

6.2.3 Variables

The main variables for the time series are annual sales and annual EBIT which refers to earnings before interest and taxes. EBIT is an accounting metric which refers to operative earnings and is a standard metric in line with generally accepted accounting principles (GAAP). The metric is of interest due to its close link to the free cash flow that a company generates.

Table 1: Descriptive Statistics (Averages) (Dataset 1 Sales time series). Source: Orbis

Statistic	**Dataset**
Ave. EBIT	10 mSEK
Ave. Revenue	91 mSEK
Ave. number of employees	52
Years of data	14

6.2.4 Outliers

Financial accounts are notorious for sometimes being unrepresentative of the true underlying financial development of a company due to periodisation, broken fiscal years, or one-off revenues, which all can heavily influence the results. Therefore, an outlier analysis was performed on the dataset to get more representative results.

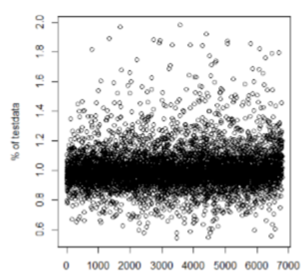

Figure 5: Plot of the distribution of previous year to the test data

See the theoretical background for the description of the method used for management of outliers and influential datapoints in the methodology section.

6.3 Dataset 2: Bankruptcy Classification

6.3.1 Description

The second model that was constructed is the logistic regression classifier. The data in this instance is not time series based and therefore the datasets will have a larger number of variables but fewer historical data points for each variable.

The dataset for default risk consists of around 30 000 limited liability companies in Sweden across a variety of industries. The dataset includes around 400 companies that have filed for bankruptcy. It is noteworthy that the dataset is highly unbalanced, which previously has been stated as a potential issue for logistic regressions.

6.3.2 Search Strategy

The search conditions that were used to generate the dataset for the bankruptcy prediction were the following:

- Operating Revenue: Minimum 30 million SEK
- Country/Region: Sweden
- Standardized Legal Form: Private limited company

The search resulted in an initial dataset of 42 791. However, this dataset is reduced to around 29 000 datapoints after removing datapoints with incomplete entries.

Table 2: Description Dataset 2 bankruptcy classifier. Source: Orbis

Statistic	Solvent	Bankruptcy
Years of Data	3	3
# of Companies	30 923	414
Mean Revenue (mSEK)	390	118
Mean EBIT (mSEK)	34	-7
Long Term Debt (mSEK)	107	3
Shareholder Equity (mSEK)	247	6
Cash (mSEK)	52	2

6.3.3 Variables

Several common financial metrics were included in the initial dataset. The full list of variables that were gathered from Orbis consists of the following:

- Revenue
- EBIT
- Long-term Debt
- Cash and cash equivalent
- Share Holder Equity

Three years of data were gathered for all the variables. The data was normalized for certain variables to increase the precision of the model. Furthermore, other regressors were constructed as different ratios of each other, such as the three-year growth rate in revenue, were used as a regressor.

6.3.4 Building Model on Unbalanced Data

Firstly, the original unbalanced data was partitioned into two groups: the training data and the test data. The training data constitutes 80% of the original dataset. The test data is built up of the remaining 20%. Secondly the training data was treated to remove outliers in the form of influential datapoints. Cook's distance was used for this purpose. The benchmark for acceptable Cooks' distance was set in accordance with the theoretical background. This resulted in 353 companies being removed from the dataset which had a significant effect on the model results. After removal of outliers and missing datapoints 23 504 datapoints remain.

6.3.5 Balancing the Dataset

The dataset for the bankruptcy prediction was heavily unbalanced with few companies entering bankruptcy compared to the total number of companies. An attempt was made to balance the data using synthetic oversampling of bankruptcies and under sampling of solvent companies. The algorithm that was used for this purpose is the ROSE package for R [Lunardon et al., 2014].

6.4 Dataset 3: Volatility

The volatility analysis is focused on averages for the historical volatility of sales and EBIT, and the incorporation of an implied market volatility component in the future volatility estimate. Data of VSTOXX was used as the forecast of the market volatility component of the future volatility.

Metric	10-year volatility	8-year (excluding pandemic)
Sales	38.8%	33.5%
EBIT	75.4%	80.1%

7 Results

In this chapter the results of the models will be presented. The topics listed below are the main sections of this chapter.

- **Prediction of Underlying Metric:** The four different models are tested in terms of their ability to predict the future values of the underlying option metric.
- **Estimation of EC Value:** The four models' evaluation in terms of their ability to accurately estimate the future pay-out from ECs with a variety of structures.
- **Bankruptcy Adjustment:** The results of the bankruptcy adjustment are presented.
- **Volatility Adjustment:** Results are presented for how the estimation of expected pay-outs were affected by incorporating implied market volatility.

7.1 Predicting Future Values for the Underlying Metric

In this section the four models (GBM, ARIMA, ANN and Hybrid) are compared in terms of their ability to accurately forecast future out of sample outcome of the underlying metrics. Each models' results were compared to the actual value and evaluated based on MSE. See the two tables below for results:

AR	**I**	**MA**	**MSE**	μ	σ
0	0	0	8.08%	0.81%	18.11%
1	0	0	3.57%	-4.10%	13.85%
0	1	0	0.94%	-2.16%	3.46%
0	0	1	2.21%	-3.41%	5.92%
1	1	0	1.49%	-2.65%	9.02%
1	0	1	4.57%	-4.22%	13.89%
0	1	1	0.87%	-1.90%	3.45%
2	0	0	8.53%	-5.60%	16.41%
0	2	0	2.37%	-3.87%	5.75%
0	0	2	1.10%	-2.44%	5.22%

Table 3: ARIMA model results showing the effect of various ARIMA model coefficients

Model	**MSE**	μ	σ
ANN	0.83%	-1.52%	3.30%
Hybrid	0.38%	-1.22%	2.32%
GBM	1.31%	2.17%	10.60%

Table 4: Showing a comparison between the MSE of the ANN model vs the Hybrid model

The results tables show forecast accuracy in terms of MSE, which in this case is defined as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{X}_T^{(i)} - \hat{\mathbf{X}}_T^{(i)})^2$$

Where n is the number of evaluated companies. The parameter μ refers to the average predicted annual growth in the underlying metric during the forecast period in the models, except for the GBM. For a company i it is defined as:

$$\mu_i = \frac{\hat{\mathbf{X}}_T^{(i)} - \mathbf{X}_t^{(i)}}{(T-t)\,\mathbf{X}_t^{(i)}}$$

It should be noted that the growth parameter is only used to give a more comparable representation of the results and is not used in the Monte Carlo simulation. However, for the GBM model the parameter μ refers to the drift of the GBM process. The parameter σ refers to the standard deviation of the forecast prediction interval, or in the case of a the GBM model it is the estimated standard deviation.

The results from the ARIMA model shows that there are some challenges to fitting advanced time series models to short time series (14 data points). However, a multitude of ARIMA models perform consistently better than a white noise process.

The results yielded that an IMA$(1,1)$ model was the best at predicting a subsequent year when compared to the other ARIMA models, yielding a MSE of 0.87%. The corresponding average growth rate and standard deviation of the model was of -1.90%, respectively 3.45%. The results show that the hybrid model is the best performing model in terms of predictions with a MSE of only 0.38% which shows that the model prediction is close to the actual value. Furthermore, the fact that the model shows an average growth rate of -1.22%, which is the closest of all models to the actual growth rate, and a low standard deviation at 2.32%, makes it the preferred choice for forecasting revenue in private companies. It is worth noting that there seems to be a bias where the models are underestimating the average growth rate at 0.2%, where the white noise process and GBM where the only models to show a positive average growth rate/drift. It should however be noted that both these models show a high standard deviation compared to the best performing models.

7.2 Model Results for Estimating the Expected Pay-out

Hereafter follows the results for each model's ability to predict the pay-out from different EC structures. The results in the tables are ordered in terms of MSE from best to worst. In this instance, MSE is defined as

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(V(\mathbf{X}_T^{(i)}) - V(\hat{\mathbf{X}}_T^{(i)}))^2$$

Model	Mean	Mean diff	MSE	CI Lower MSE (95%)	CI Upper MSE (95%)
Actual	0.0363	-	-	-	-
Hybrid	0.0488	0.0125	0.0206	0.0148	0.0263
010	0.0596	0.0233	0.0229	0.0162	0.0296
Auto	0.0563	0.0199	0.0240	0.0168	0.0313
002	0.0586	0.0223	0.0263	0.0180	0.0346
ANN	0.0539	0.0175	0.0267	0.0182	0.0353
GBM	0.0934	0.0594	0.0292	0.0033	0.0525
011	0.0617	0.0253	0.0433	0.0258	0.0608
012	0.1090	0.0727	0.0640	0.0336	0.0944

Table 5: Linear pay-out structure results

The results show that a linear pay-out structure has the hybrid model the best performing model. It is statistically ($p = 0.05$) better than the MA(2), ANN, GBM, IMA$(1, 1)$ and IMA$(1, 2)$ in terms of MSE, but not than the Auto-arima and I(1) model. Furthermore, the variance is lower for the hybrid model compared to the other models. The performance of the model is deemed to be satisfactory as the mean value of the predicted pay-out deviates from the actual value with around 30%, which is deemed decent when considering that no other qualitative input or adjustments have been made on each case, which usually happens during the acquisition process.

Model	Mean	Mean diff	MSE	CI Lower MSE (95%)	CI Upper MSE (95%)
Actual	0.3000	-	-	-	-
GBM	0.2480	-0.0521	0.2246	-0.2731	0.7224
Hybrid	0.1451	-0.1549	0.3099	-0.0149	0.6348
ANN	0.1532	-0.1468	0.3102	-0.0165	0.6369
Auto	0.1680	-0.1321	0.3134	-0.0208	0.6477
002	0.1631	-0.1369	0.3156	-0.0214	0.6525
010	0.1804	-0.1196	0.3156	-0.0240	0.6552
011	0.1722	-0.1278	0.3157	-0.0229	0.6544
012	0.2359	-0.0641	0.3315	-0.0403	0.7034

Table 6: Event structure

Results for the event-based pay-out structure shows that the GBM model yields the best results. However, the mean squared error, and corresponding confidence intervals makes it impossible to distinguish the different models statistically. This is since the actual pay-out takes on values as 0 or 1, while each model gives values between 0 and 1. Thereby, the difference becomes higher, and thereby MSE can be seen as flawed when it comes to valuing this type of contract.

Model	Mean	Mean diff	MSE	CI Lower MSE (95%)	CI Upper MSE (95%)
Actual	0.0709	-	-	-	-
Hybrid	0.0609	-0.0100	0.0267	0.0182	0.0353
ANN	0.0706	-0.0004	0.0283	0.0189	0.0376
Auto	0.0679	-0.0030	0.0295	0.0196	0.0394
002	0.0642	-0.0067	0.0301	0.0199	0.0404
010	0.0711	0.0002	0.0324	0.0210	0.0438
GBM	0.0981	0.0249	0.0328	0.0007	0.0649
011	0.0694	-0.0015	0.0367	0.0229	0.0505
012	0.1193	0.0484	0.0672	0.0337	0.1008

Table 7: Consecutive growth EC (3) in section 5.4.1

Lastly, the hybrid model once again outperforms the other models on the path dependent EC structure with consecutive growth. Both the mean squared error and the mean difference is low, indicating accurate valuations. When observing the confidence intervals for MSE, there is a significantly better performance from the Hybrid model compared to the IMA$(1, 1)$ and IMA$(1, 2)$ models. However not to the rest of the models. Once again, the variance of the hybrid model is the lowest of them all.

7.3 Bankruptcy Adjustment and Logistic Regression

The intent described in the estimation methodology was to use the prediction of future bankruptcy to adjust the estimation of expected EC pay-out for the four different models. Being able to perform this adjustment requires sufficiently accurate predictions of the future bankruptcy probability for each company in the test data. First, a logistic regression model was constructed on the initial unbalanced dataset. However, it was found that the model was not able to predict bankruptcies given the unbalanced nature of the dataset. Secondly, a logistic model was constructed on the dataset after it had been balanced using synthetic sampling combined with under sampling of the data. The model was now able to perform prediction, however it was unable to reach sufficient accuracy.

Unfortunately, the dataset was highly unbalanced which made it difficult to achieve reasonable prediction accuracy for the regression model given the method that was chosen. As a result, the bankruptcy adjustment was not used for adjusting the expected pay-outs as was intended in the estimation background due to the low accuracy of the model. Thus, it was not possible to adjust for bankruptcies in this case to improve the estimation of the future value of the underlying option metric.

Below are the model results for the logistic regression models on the balanced and unbalanced datasets respectively.

7.3.1 Logistic Regression on Unbalanced Dataset

Below are the model coefficients for the logistic regression performed on the initial unbalanced dataset. This means that there is a larger number of companies that do not default compared to the number of companies did default.

Variable	Log Odds	Odds	p-value	significance
EBIT	-0.011	0.989	3.50e-12	***
Cash and equivalents	-0.131	0.877	1.77e-12	***
Shareholder equity	-0.006	0.995	2.23e-05	***
CAGR Revenue 3 years	0.085	1.089	0.0425	*
Cash and equivalents / Revenue	-0.674	0.509	0.1696	

Table 8: Model coefficients for the unbalanced dataset logistic regression

The significance of the model can be approximated through considering the null deviance in relation to the residual deviance. A generalized R^2 value can be calculated as follows:

$$R^2 = 1 - \frac{\text{residual deviance of fitted model}}{\text{null deviance without fitted model}} = 1 - 0.908 = 0.091$$

A R^2 value of 1 would indicate that the model perfectly explains the data. An R^2 of 0 indicates that the model contributes nothing to the prediction accuracy. As a result, an R^2 value of 0.091 indicates that the model might be able to explain some of the variation, although only a small part. Thereafter, the model's ability to predict bankruptcies within the test dataset was evaluated. A confusion matrix was generated which can be seen below:

	Actual Solvent	Actual Bankruptcy
Predicted as Solvent	5847	70
Predicted as Bankruptcy	1	0

Table 9: Confusion matrix of the prediction model accuracy on the unbalanced dataset

From the confusion matrix the predictive ability of the model on the unbalanced dataset is not significant as it only predicted one bankruptcy and this bankruptcy was a solvent company.

7.3.2 Logistic Regression on Balanced Dataset

The table below shows the model coefficients for the logistic regression model for bankruptcy prediction on the balanced data.

Variable	Log Odds	Odds	p value	significance
Revenue	-3.575e-05	0.9999	1.18e-06	***
EBIT	-3.701e-04	0.9996	9.86e-11	***
Long term Debt	-9.031e-05	0.9999	4.19e-06	***
Cash and Cash Equivalents	-2.367e-04	0.9998	1.91e-08	***
Shareholder Equity	-2.497e-05	0.9999	0.000841	***
CAGR Revenue 3 Years	8.607e-02	1.0898	6.45e-07	***
Long Term Debt / Shareholder Equity	-7.098e-03	0.9929	0.048915	*
EBIT / Revenue	-1.487e-01	0.8618	2e-16	***
Long Term Debt / Revenue	-6.773e-03	0.9932	0.007807	**
Shareholder Equity / Revenue	-1.449e-03	0.9986	0.000661	***

Table 10: Final coefficients for the logistic regression on the balanced dataset

When calculating the generalized R^2 value for the logistic regression model on the balanced dataset. The value is as follows:

$$R^2 = 1 - \frac{\text{residual deviance of fitted model}}{\text{null deviance without fitted model}} = 1 - 0.988 = 0.01$$

The generalized R^2 value is only 1% which means that the model barely brings any benefit to the balanced dataset it has been trained on. When the model is instead evaluated in terms of its prediction ability on test data the following confusion matrix is generated:

	Actual Solvent	Actual Bankruptcy
Predicted as Solvent	1603	1262
Predicted as Bankruptcy	1342	1672

Table 11: Confusion matrix for the predictions on the balanced dataset

Compared to the previous model that could not catch any bankruptcies, this model on a balanced dataset is able to correctly predict 1672 bankruptcies and to predict 1603 correctly solvent companies. However, given the 2604 wrong predictions, the models predictions are attributed to guessing randomly. Therefore, no runs with Monte Carlo valuation were performed as it was deemed that it would not contribute to more accurate valuations.

7.4 Volatility Analysis Results

The annualized implied volatility of the VSTOXX index was on average 18.95% during the three months before the start of 2018. Thereby the squared volatility for each model that was used is:

$$\sigma_{tot}^2 = \sigma_{model}^2(1-\alpha) + \alpha\sigma_{VSTOXX}^2 = \sigma_{model}^2(1-\alpha) + \alpha(18.95\%)^2$$

The models were evaluated on the scenario with consecutive growth and different levels of α. When analysing the results for individual models, there are patterns that shows to which degree it is preferable to use implied volatility in the model.

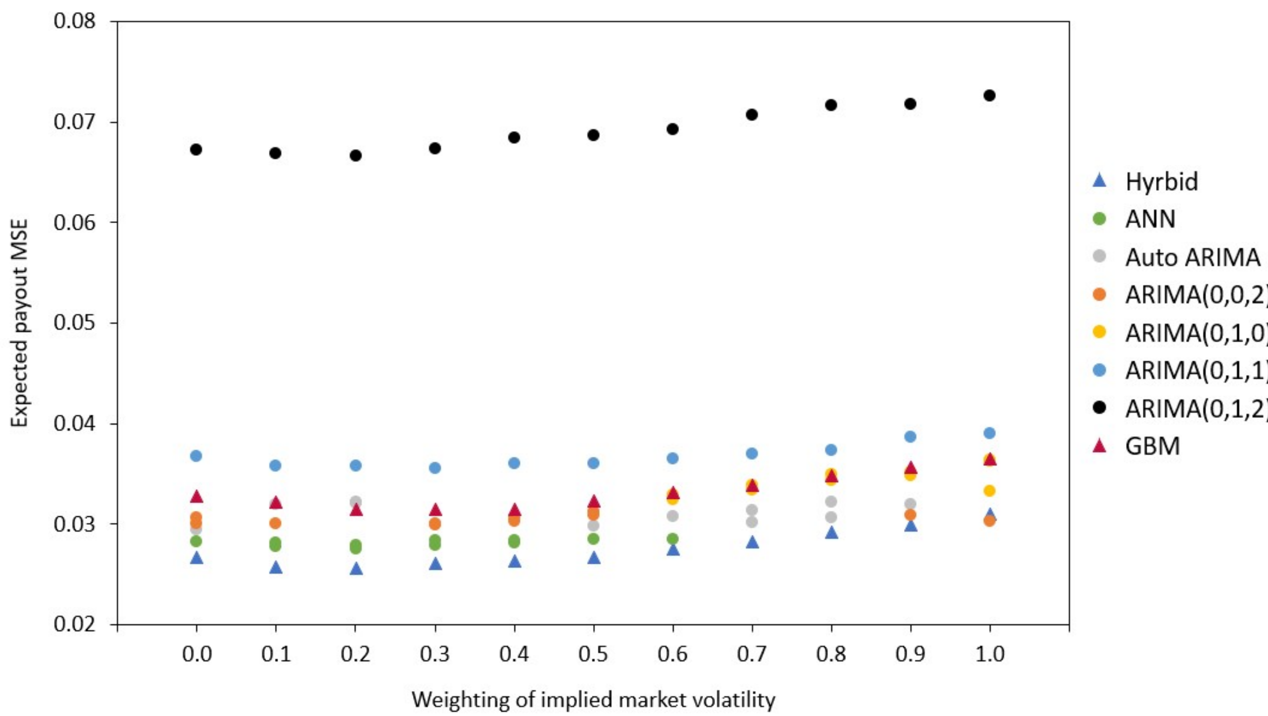

Figure 6: Plot showing how the expected pay-out MSE for different α

When the Monte Carlo simulation is run, using the formula above for the future volatility and varying the value of α, the plot shown shows the corresponding MSE of expected pay-outs. A combination of the company forecast volatility and the market volatility seem to generate better estimates. The most accurate valuation in terms of MSE seem to be given by an α of 0.3 and the Hybrid model. There seems to be a pattern with results improving when incorporating around 20-30% of implied volatility. However, when analysing the statistical significance of the results ($p = 0.05$), the conclusions are not as clear. Monte Carlo simulation with the hybrid model was run while varying the parameter α. In other words, varying how much weight the implied market volatility has in the simulation of expected pay-outs. The confidence intervals of the MSEs for different levels of α was used to test if incorporating various levels of α generated significantly better MSE estimations. The results are presented in Table 12 below. The MSE of the hybrid model with α_1 was compared with the MSE using the weighting α_2. If the MSE from using α_1 was significantly better than the corresponding MSE with α_2 it is denoted with 1 in the table. If the reverse holds true, it is denoted as -1. Otherwise, it is denoted by 0. The table below shows that incorporating no market volatility resulted in a significantly better MSE estimate at ($p = 0.05$) than incorporating only the estimate of market volatility.

α_2 \ α_1	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0											
0.1	0										
0.2	0	0									
0.3	0	0	0								
0.4	0	0	0	0							
0.5	0	0	0	0	0						
0.6	0	0	0	0	0	0					
0.7	0	0	0	0	0	0	0				
0.8	0	1	0	0	0	0	0	0			
0.9	0	1	1	1	1	0	0	0	0		
1	1	1	1	1	1	1	1	0	0	0	

Table 12: Results of implied volatility weighting for the hybrid model

Similar analysis of statistically significant improvement in MSE was performed for all models tested for expected pay-out prediction. The results are shown in Table 13. Here each cell shows a positive number to indicate the number of models where this level of α_1 had an MSE that was significantly better than α_2. The following negative number in the cell shows the number of models where this level of α_1 produced significantly worse MSE than α_2. The results in the table show that there is no clear level of α that across models generate significantly improved MSE for expected pay-out forecasts. For instance, using the models own volatility makes the valuation statistically better in four cases, while statistically worse in three. Furthermore, there is no value on alpha that makes the models in general perform better than the model's own estimates of volatility. On the contrary, the results lean towards the models own volatility estimates being generally better than using the implied volatility.

α_2 \ α_1	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0	0/0										
0.1	2/-1	0/0									
0.2	2/-1	1/-1	0/0								
0.3	1/-1	1/-1	1/-1	0/0							
0.4	1/-1	1/-1	1/-1	0/0	0/0						
0.5	0/0	1/-2	1/-2	2/-1	1/-1	0/0					
0.6	1/-1	1/-2	1/-2	2/-2	2/-2	1/-1	0/0				
0.7	2/-2	2/-2	1/-1	2/-2	2/-2	2/-2	1/-1	0/0			
0.8	3/-2	4/-2	2/-1	3/-2	2/-2	2/-2	1/-1	0/0	0/0		
0.9	4/-2	6/-1	6/-2	5/-2	4/-2	3/-2	2/-1	1/-1	1/-1	0/0	
1	5/-3	5/-2	6/-2	4/-2	4/-2	3/-3	4/-2	2/-1	2/-1	1/-1	0/0

Table 13: Aggregated implied volatility results

8 Discussion

8.1 Data Limitations

The purpose of the thesis was primarily to test various quantitative methods of estimating the expected pay-out from EC contracts. The main challenge of this pursuit is that ECs are used for M&A transactions of private companies, which in turn often have limited availability of financial data. The length of time series data for annual financial metrics ranged between 13-14 years for the companies. There is no clear boundary for how long time series should be for time series models to efficiently forecast future values. This is especially the case when working with annual data. If the method works or not depends on how much randomness is in the data and how many parameters are being fitted in the model, since there must be more datapoints than parameters. The results found in this thesis indicate that it is possible to use both time series models and ANN models on sales figures to perform forecasts of the commonly used underlying metrics for ECs. Especially sales data seem to have low enough randomness for time series and ANN models to work moderately well despite little available data.

8.2 Forecasting of the Underlying Metric

The first test evaluated how accurate GBM, ARIMA, ANN and Hybrid models were at predicting the future values of the underlying metric of the EC. The result showed that the time series models performed well when predicting future values. The best performing model turned out to consistently be the Hybrid model, although it cannot be statistically confirmed that it outperforms all other models. The suggested explanation is that the hybrid model can pick up both linear and non-linear patterns and should thus be able to produce better predictions. Thereby, it can extract more information from the otherwise limited dataset, the length of which is challenging for the other models. An interesting note is that the GBM model performed well in terms of mean prediction although with high volatility leading to higher MSE than many other models.

8.3 Expected Pay-out Estimation

Monte Carlo simulation was used together with the four different forecast models for three different contract structures, linear option-based, event-based, and consecutive growth pay-out. The results showed that the hybrid model was the most accurate model. A reason is the fact that the model has the lowest error-rate when predicting the underlying metrics growth, while simultaneously having the lowest volatility of all models. However, for event-based ECs the GBM model had the lowest MSE and showed the best results, with the hybrid model as runner up. Furthermore, it is noteworthy that the ANN is performing top three in two of the three cases. This further strengthens the notion that the hybrid model is the strongest of the lot, as it is based on the ANN but can extract more information from the dataset. The ARIMA-models are in general producing decent results, but worse than the ANN and hybrid model. An explanation could be that non-linear aspects could remain in the residual of the ARIMA models when estimating the parameters of the EC.

8.4 Bankruptcy Prediction

In this thesis a hypothesis was posed that the estimation of the expected pay-out can be improved by accounting for the risk of bankruptcy. In other words, that the company which was bought enters bankruptcy and the underlying metric of the options goes to zero. An attempt was made to build a model for bankruptcy prediction based on logistic regression. The intention was to then use the bankruptcy predictions to adjust the expected pay-out estimates.

Firstly, a logistic regression model on the original unbalanced dataset that included a large majority of companies that had not gone into bankruptcy and a small minority of companies that went into bankruptcy. This model received a somewhat significant generalized R^2 value at 9.1%, and with significant coefficients. However, it did not successfully predict bankruptcies on the test data which was shown by the confusion matrix. A hypothesis is that the model was overfitted due to the unbalanced nature of the dataset.

Secondly, the model based on the balanced dataset showed results of a generalized R^2 close to zero, with the coefficients having no impact on the odds function. Furthermore, the confusion matrix did not yield any definitive results, with the model practically guessing randomly. Since the prediction accuracy was not sufficiently high, bankruptcy prediction was not used for adjusting the expected pay-out as it was deemed that it would not yield any improvements to the model. However, since previous studies have shown significant results for attempting to predict bankruptcies with similar methodologies, it could be of interest to make another attempt on those datasets and models in the future.

As the model did not accurate results, an alternative approach could have been to to use a fixed bankruptcy probability which would be the same for all companies. This probability could be computed as the average bankruptcy rate based on e.g., company size or industry sector. This solution would have some flaws in individual cases but could provide a rough solution to the problem at hand.

8.5 Implied Volatility

Volatility is an essential parameter to understand with regards to the value of an option, which is thus also true for the value of ECs. The initial approach used in this thesis was to use the prediction volatility from the time series estimation of the underlying metric as the future volatility. However, this only incorporates the historical volatility in the time series and the uncertainty in the prediction. A secondary approach was tested where a linear combination of a measure of the future market volatility and the prediction volatility was weighted together. In other words, the implied market volatility for the period for the VSTOXX 50 index was used in combination with the prediction volatility. Including market volatility showed promising results for some different models and weights. However, there was no clear weighting of the market volatility that proved statistically better across all models. Therefore, the usefulness of implied volatility can neither be confirmed or dismissed by these results. Therefore, further analysis on this matter will be needed in the future. A positive aspect however is that the general results seem to lean in favour of the models own volatility rather than only on the general markets implied volatility, which is in line with expected results. It would be interesting to test using the implied volatility from listed options instead of the implied volatility from a market index. But finding high quality data for historical options prices proved difficult.

8.6 Black Scholes vs Monte Carlo

Throughout this article most of the focus has been directed towards Monte Carlo methods for valuing ECs. One might ask why the focus is directed towards Monte Carlo simulation and not Black Scholes. The Black Scholes model has its benefits since one can value options analytically. However, there are also limitations. The assumptions of the Black Scholes model become hard to justify for real options where the underlying is not listed on an arbitrage free market, the formula is therefore often miss-used in the industry. Furthermore, there is the issue with path dependent options that are not possible to evaluate when using Black-Scholes. Monte Carlo simulation on the other hand is not limited to the strict assumptions of the Black Scholes model, the distribution function of the underlying metric can be adjusted, and simulations can be made for a wider variety of option structures. Moreover, it is possible to estimate the expected pay-out separately from the discounting of the pay-out which was suitable for this thesis. The main drawback is that many simulations are required for Monte Carlo simulations to converge. However, since ECs are often valued for one company at a time, and since the simulation is generally done with discrete time-steps the computational requirements should not pose a large problem in practice. Based on what has been uncovered through this thesis, Monte Carlo seems to hold the most benefits and promise for future development of EC valuation due to more solid assumptions and flexibility.

9 Conclusions

This thesis has sought to answer how to estimate the pay-out from ECs. Four models have been tested for this purpose. A model based on Geometric Brownian Motion, an ARIMA model, an ANN model, and a hybrid model between an ARIMA and ANN.

Firstly, it is concluded that time series models can be utilized successfully to predict future values for the most common underlying metrics of ECs. The model that performed the most accurate results was the hybrid model. There were simple models that performed better in terms of the mean value, however these models had higher volatility. Secondly, the forecasts using the four different models were used to estimate the expected pay-out of ECs with three different structures. It was found that the hybrid model performed well consistently. However, it was not the best model for all contract structures as the GBM model was the most accurate for event-based structures. Thirdly, the logistic regression model that was trained for the purpose of bankruptcy prediction did not reach sufficient accuracy for bankruptcy adjustment to be incorporated in the estimation of expected pay-out. Thereby it is concluded that it cannot be confirmed as a useful tool for EC valuation from this thesis. Furthermore, an attempt was made to reach a more accurate estimate by using a linear combination of the forecast volatility from the model and the implied volatility of a European options index. The result showed some promising results with estimations that included a market volatility which at times performed better than with only the models own suggested volatility. However, no specific weighting could be said to be consistently better with statistical significance. Therefore, further research in this area is recommended. Lastly, the thesis suggests that Monte Carlo simulation is the preferred model for valuing ECs. The main benefits being flexibility with regards to the distribution used and the possibility to model a large variety of option structures. Furthermore, computational requirements for Monte Carlo simulations are not deemed to be an issue when valuing ECs in practice.

10 Areas for Future Research

Given the limited research that has been done within the field there remain a lot of areas for future research to be done. This thesis has touched on several topics that deserves a deeper exploration in the future. Some suggestions of such areas are as follows:

- **Volatility** in the underlying option metric is essential to the value of the option. More research is needed on the volatility behaviour of various financial metrics on private companies. Furthermore, a potentially interesting area to look deeper into is how the implied future market volatility and the company specific volatility forecast can be jointly used to forecast future volatility in financial metrics, based on a complete dataset with historical option prices with long maturities.
- **Monthly/quarterly data set** as a limiting factor in this study has been the short time series. To study financial metrics with more randomness than sales or to test prediction further into the future, more data is desirable. A solution would be to experiment with monthly or quarterly data for public companies, or if having access to a more through dataset with private companies.
- **Bankruptcy Prediction** did not reach sufficient predictive ability in this thesis. However, given previous research it seems possible to reach some predictive accuracy with alternative coefficient estimation methods and sampling techniques.
- **Joint Subjective and Objective Estimation** could be promising and solve some challenges with objective estimation of parameters for EC valuation with quantitative methods. It might be interesting to analyse approaches for incorporating both subjective expert opinion and objective estimation for selecting parameters for EC valuation.

Valuing ECs is no simple task, and this thesis has highlighted both some of the challenges and opportunities for future progress. These areas above should provide further insight into the field if pursued and thoroughly researched. Given the large increase in the use of ECs during acquisitions and the increasing accounting requirements on ECs the need for accurate valuation is increasing year by year. Further research on the topic would provide value to the industry and make the use of ECs more efficient as a financial negotiation tool during acquisitions.

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11 Appendix

11.1 Pseudo-code for Expected Pay-out Calculations

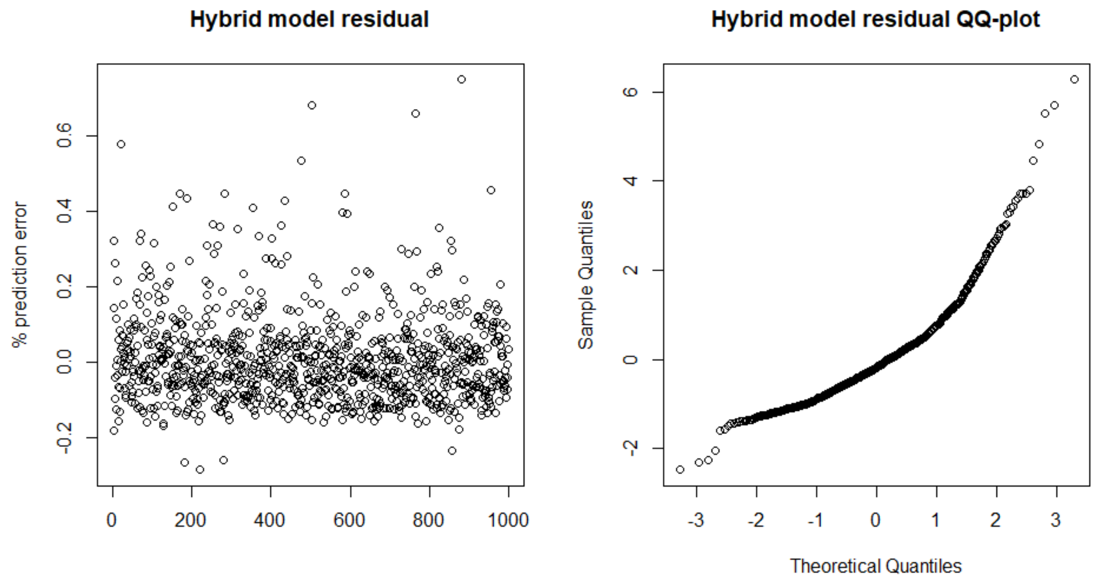
Let T be the last year of the time series and h be the number of years to be forecasted. Furthermore, to generalize if there would be a path dependent EC, in the description below V is allowed to take either a single variable, or a vector of multiple forecasts which is then denoted as $\vec{\hat{\mathbf{X}}}_T$. A pseudo-code of the Monte Carlo valuation algorithm used goes as follows:

1. Compute the coefficients for the ARIMA model based on $\{\mathbf{X}_t^{(i)}\}_{T-h\geq t>0}$
2. Use the model to forecast h lags into the future, i.e., $\vec{\hat{\mathbf{X}}}_T^{(i)} = \begin{bmatrix} \hat{\mathbf{X}}_{T-h+1}^{(i)} \\ \vdots \\ \hat{\mathbf{X}}_T^{(i)} \end{bmatrix}$
3. Compute $\hat{\sigma}^{(i)}$
4. Compute $V(\vec{\mathbf{X}}_T^{(i)})$
5. Draw new $\vec{\hat{\mathbf{X}}}_T^{(i),j}$ for $j = 1 \ldots N$
6. Compute $V(\vec{\hat{\mathbf{X}}}_T^{(i),j})$ for all j
7. Compute $\epsilon^{(i),j} = V(\vec{\hat{\mathbf{X}}}_T^{(i),j}) - V(\vec{\mathbf{X}}_T^{(i)})$ for all j
8. Compute $MSE^{(i)} = \frac{1}{N}\sum_{j=1}^{N}(\epsilon^{(i),j})^2$
9. Repeat for each company i

where i is an index for a company. Note that the methodology likewise holds for the GBM, ANN and Hybrid model, with minor changes for how the forecast and computation of $\hat{\sigma}^{(i)}$ is performed.

11.2 Residual Analysis for Drift Prediction

Analysis of the residuals of the prediction models, when assuming a normal-distribution of the underlying metric, show that the Quantile-Quantile-plot (QQ-plot) does not form a straight line and that a normal distribution is not an accurate fit for the data. When assuming that $\mathbf{X}$ has a log-normal distribution, the QQ-plot shows that the residuals form a straight line, which thereby confirms the assumption that $\mathbf{X}$ is log-normally distributed. Furthermore, it is noteworthy that there are datapoints that the model still has issues with predicting. These could potentially be considered as outliers, but they have been included to keep the data representative.



(a) (b)

Figure 7: (a) Distribution of the difference between the hybrid models predicted value and the actual value. (b) Quantile-Quantile plot of hybrid models residual with normal distribution assumption

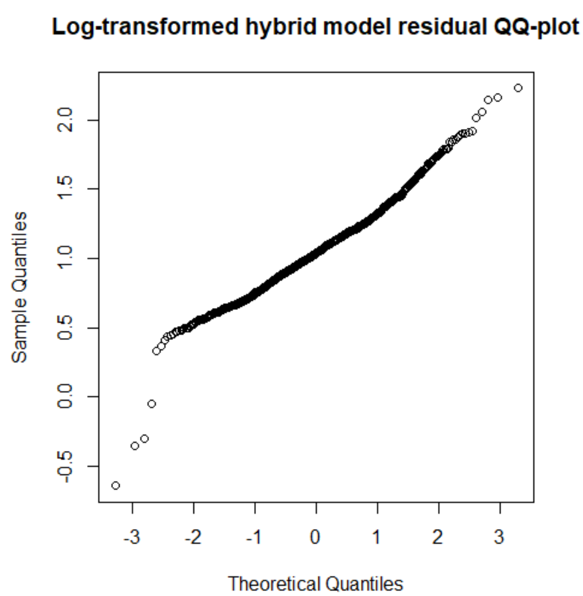


Figure 8: QQ-plot of hybrid model residual with log-normal distribution assumption.

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